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Inducible AAV system comprising cumate operator sequences

Abstract

The invention relates to nucleic acid molecules, vectors and plasmids comprising AAV cap genes and rep genes, wherein the cap and rep genes are both operably-associated with an inducible promoter which comprises one or more cumate operator (CuO) sequences. The invention further relates to producer and packaging cell lines which are useful in the production of Adeno-Associated Virus (AAV) particles.

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Background/Summary

CROSS-REFERENCE

(1) This application is a 371 U.S. national phase of PCT/GB2020/050252, filed Feb. 5, 2020, which claims priority from GB 1901571.8, filed Feb. 5, 2019, both which are incorporated by reference in its entirety.

SEQUENCE LISTING STATEMENT

(2) This application includes an electronically submitted Sequence Listing in text format. The text file contains a sequence listing entitled "21-0965-WO-US_Sequence-Listing_ST25.txt" created on Jan. 5, 2022 and is 52 kilobytes in size. The Sequence Listing contained in this text file is part of the specification and is hereby incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

(3) The invention relates to nucleic acid molecules, vectors and plasmids comprising AAV cap genes and rep genes, wherein the cap and rep genes are both operably-associated with an inducible promoter which comprises one or more cumate operator (CuO) sequences. The invention further relates to producer and packaging cell lines which are useful in the production of Adeno-Associated Virus (AAV) particles.

BACKGROUND OF THE INVENTION

(4) AAV vectors are single-stranded DNA viruses that belong to the Parvoviridae family. This virus is capable of infecting a broad range of host cells, including both dividing and non-dividing cells. In addition, it is a non-pathogenic virus that generates only a limited immune response in most patients.

(5) The native AAV genome comprises two genes each encoding multiple open reading frames (ORFs): the rep gene encodes non-structural proteins that are required for the AAV life-cycle and site-specific integration of the viral genome; and the cap gene encodes the structural capsid proteins.

(6) In addition, these two genes are flanked by inverted terminal repeat (ITR) sequences consisting of 145 bases that have the ability to form hairpin structures. These hairpin sequences are required for the primase-independent synthesis of a second DNA strand and the integration of the viral DNA into the host cell genome.

(7) In order to eliminate any integrative capacity of the virus, recombinant AAV vectors remove rep and cap from the DNA of the viral genome. To produce such vectors, the desired transgene(s), together with a promoter(s) to drive transcription of the transgene(s), is inserted between the inverted terminal repeats (ITRs); and the rep and cap genes are provided in trans in a second plasmid. A third plasmid, providing helper genes such as adenovirus E4, E2a and VA genes, is also used. All three plasmids are then transfected into cultured mammalian cells, such as HEK293.

(8) Over the last few years, AAV vectors have emerged as an extremely useful and promising mode of gene delivery. This is owing to the following properties of these vectors: AAVs are small, non-enveloped viruses and they have only two native genes (rep and cap). Thus they can be easily manipulated to develop vectors for different gene therapies. AAV particles are not easily degraded by shear forces, enzymes or solvents. This facilitates easy purification and final formulation of these viral vectors. AAVs are non-pathogenic and have a low immunogenicity. The use of these vectors further reduces the risk of adverse inflammatory reactions. Unlike other viral vectors, such as lentivirus, herpes virus and adenovirus, AAVs are harmless and are not thought to be responsible for causing any human disease. Genetic sequences up to approximately 4500 bp can be delivered into a patient using AAV vectors. Whilst wild-type AAV vectors have been shown to sometimes insert genetic material into human chromosome 19, this property is generally eliminated from most AAV vectors by removing rep and cap genes from the viral genome. In such cases, the virus remains in an episomal form within the host cells. These episomes remain intact in non-dividing cells, while in dividing cells they are lost during cell division.

(9) In addition to being involved in AAV DNA replication and gene regulation, the Rep proteins are known to have cytostatic and cytotoxic effects on cells which express them.

(10) HEK293 cell lines expressing Rep proteins typically grow more slowly, with colony-forming

efficiency reduced by over 90% when cell lines express Rep 78. The cells look morphologically typical, but are increasingly found in S-phase, suggesting they do not introduce a cell cycle block but slow or inhibit the cell cycle.

(11) The cytostatic and cytopathic effects of the large Rep proteins have the potential to be deleterious to a stable cell line expressing AAV packaging genes if expressed constitutively. In the native configuration, the large Rep proteins are expressed from the p5 promoter, which is itself activated by the large Rep proteins and adenoviral helper genes. The small Rep proteins, Rep 52 and Rep 40, are expressed from the p19 promoter, which is found within the rep gene and is activated by the larger Rep proteins.

SUMMARY OF THE INVENTION

(12) The invention relates to nucleic acid molecules, vectors and plasmids comprising AAV cap genes and rep genes, wherein the cap and rep genes are both operably-associated with an inducible promoter which comprises one or more cumate operator (CuO) sequences. The invention further relates to producer and packaging cell lines which are useful in the production of Adeno-Associated Virus (AAV) particles.

(13) The inventors have recognised that constitutive expression of the rep genes in cell lines is undesirable and that it would be advantageous to control expression of the rep genes from an inducible promoter. Furthermore, induction should allow sufficient rep gene expression to allow for the production of high titres of AAV.

(14) A very wide variety of methods exist to regulate gene expression in mammalian cells using inducible systems. Examples include the Tet and lac operon-based systems, riboswitches, Cre-loxP/FLP-FRT (and conditional variants) and light and temperature sensitive regulators. Transcriptional activators may also be constructed using modified forms of gene editing technologies such as CRIPSR/Cas9, Zinc-Fingers, and other nuclease-deficient DNA targeting mechanisms.

(15) It is widely known in the art that promoters can be modified to contain DNA sequences targeted by pre-existing DNA targeting proteins, or can be specifically targetted using programmable methods (e.g. CRISPR) (e.g. Maximilian Hörner, Wilfried Weber, Molecular switches in animal cells, FEBS Letters, Volume 586, Issue 15, 2012, pages 2084-2096).

(16) After testing a number of different systems and promoters using reporter genes and in AAV production assays, the inventors have found that a system based around a minimal promoter which comprises one or more cumate operator sites (CuOs) provides a high level of control over AAV gene transcription.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. **1**A. Schematic diagram of dual cumate control system in the absence of cumate where the target regulated gene is transcriptionally off.

(2) FIG. **1**B. Schematic diagram of dual cumate control system where addition of cumate relieves all modes of repression allowing activation of target regulated gene.

(3) FIG. **2** shows the organisation of the Rep and Cap protein genes in the wild-type AAV genome.

(4) FIG. **3**A. Position of CuO site in enhancer region (SEQ ID No: 21, and reverse complement is SEQ ID NO:23).

(5) FIG. **3**B. Positioning of second downstream CuO relative to transcription start site (SEQ ID No: 22, and reverse complement is SEQ ID NO:24).

(6) FIG. **4**. Mean fluorescence intensity of HEK293 cells transfected with CymR and cumate regulated eGFP constructs. Error bars depict standard deviation (n=3, biological triplicate, analytical single).

(7) FIG. **5**. Mean fluorescence intensity of HEK 293 cells. Error bars depict standard deviation (n=3, biological triplicate, analytical single). X-axis depicts the additional plasmid added (CymR-TAD or stuffer OG10).

(8) FIG. **6**. Mean fluorescence intensity of HEK 293 cells in additional experiment. Error bars depict standard deviation (n=3, biological triplicate, analytical single). X-axis depicts the additional plasmid added (CymR-TAD or stuffer OG10).

(9) FIG. **7**. qPCR data showing rAAV5 vector genomes per mL from the first rAAV production experiment. Error bars depict standard deviation, with n=3 (biological triplicate, analytical single).

(10) FIG. **8**. Cell specific productivity from first rAAV production experiment, calculated using cell count at time of transfection, and viral genomes per mL. Error bars depict standard deviation, with n=3 (biological triplicate, analytical single).
(11) FIG. **9**. Mean fluorescence intensity of HEK293 cells. Error bars depict standard deviation, n=3 (biological triplicate, analytical single).
(12) FIG. **10**A. Flow cytometry data from eGFP expression using dual cumate system−mean fluorescence intensity. Error bars depict standard deviation (n=3, biological triplicate, analytical single).
(13) FIG. **10**B. Flow cytometry data from eGFP expression using dual cumate system−median fluorescence intensity. Error bars depict standard deviation (n=3, biological triplicate, analytical single).
(14) FIG. **11**. qPCR data showing vector genomes per ml (VG/mL).
(15) FIG. **12**. qPCR data showing vector genomes per ml (VG/mL).
(16) FIG. **13**. qPCR data showing rAV5 vector genomes per mL. Error bars depict standard deviation, with n=4 (biological duplicate, analytical duplicate).

DETAILED DESCRIPTION OF THE INVENTION

(17) Cumate operator sites (abbreviated herein as "CuOs") are DNA elements which are capable of being bound by a cumate repressor protein (CymR). The CuO is therefore a DNA recognition sequence for CymRs.
(18) In the absence of the cumate small molecule, the cumate-repressor protein (CymR) binds to cumate operator sites (CuO) within an associated promoter. Where the operator sites are proximal to the minimal/core promoter region, the CymR protein sterically blocks formation of the transcription initiation complex and thus prevents transcription.
(19) In the presence of cumate, the CymR protein undergoes a conformational change, altering its DNA-binding properties and reducing its affinity for the CuO sites. Thus in the presence of cumate the repression of promoter activity via CymR binding is relieved and transcription proceeds.
(20) 4-Isopropylbenzoic Acid (Cumate)
(21) ##STR00001##
(22) Uses of CuO sites and the CymR protein are known in the art (e.g. "High-level recombinant protein production in CHO cells using lentiviral vectors and the cumate gene-switch", Gaillet B, et al. Biotechnol. Bioeng. 2010 Jun. 1; 106(2):203-15. doi: 10.1002/bit.22698).
(23) Cumate-based systems have not, however, previously been used in the context of the regulation of AAV genes. This is in part because the genetics of AAV are complex, with all existing systems using the native promoters found within the AAV virus as it is in nature to drive Rep and Cap expression. The inventors determined that it was possible to change the expression context of both Rep and Cap such that they were controlled by a single polymerase II promoter to drive expression of the main AAV genes. Only once this had been achieved did it then become apparent that the expression of these genes could be regulated using an inducible system such as when using Cumate.
(24) The invention does not, however, use cumate in the standard manner, i.e. to relieve repression due to the binding of a CymR polypeptide to a CuO site.
(25) In one aspect, the invention relates to a packaging cell line wherein the AAV cap and rep genes—in reverse order compared to their wild type configuration—are placed under the control of a minimal promoter which comprises one or more CuO sites. Placing the cap and rep genes in this reverse order has the advantage of optimising the ratios and amounts of Rep and Cap proteins which are produced during the AAV vector production process. Induction of cap and rep expression, as a single transcript, is then achieved by induction with a CymR-transcriptional activator domain polypeptide. This may be provided from a plasmid which is transfected into the cell.
(26) In a further aspect, the invention relates to a complex dual-control system for controlling expression of AAV genes in a producer cell line. The producer cell line comprises a number of DNA constructs integrated into its genome: The first DNA construct comprises cap and rep genes under the control of a minimal promoter, wherein the promoter comprises one or more CuO sites. The second construct comprises a gene encoding a reverse cumate repressor (rcCymR) polypeptide linked to a transcriptional activator domain under the control of a promoter which comprises at least one CuO site. The third DNA construct comprises a gene encoding a CymR polypeptide under the control of a constitutive promoter. In this dual control aspect, in the absence of cumate, the CymR binds to the promoter region of the rcCymR

gene and represses its transcription; the CymR also binds to the minimal promoters driving the cap-rep (and/or Adenoviral) genes, sterically blocking any residual rcCymR-TAD binding. In the presence of cumate, the repression of rcCymR-TAD transcription is relieved, as is the steric blocking, and the rcCymR-TAD is allowed to bind to the promoter of the cap-rep genes (and/or Adenoviral genes), and activates transcription—resulting in rAAV production. Cumate may be introduced into the cell by adding cumate into the media surrounding the cell, at the desired time. This aspect is illustrated in FIGS. **1**A and **1**B.

(27) In addition to regulating expression of the Rep and Cap proteins, the same systems may be used to regulate the expression of Adenoviral proteins (e.g. E4) which may also be present in the packaging or producer cell lines, and which may have cytotoxic or cytostatic effects.

(28) The invention also provides nucleic acid constructs for use with the cell packaging and producer cell lines.

(29) The use of a cumate-based system has particular advantages over antibiotics-based systems in that the possibility of contaminating antibiotics being a carried over into the final AAV preparations is removed. Furthermore, in the context described here, the system allows for a genetic-based induction system in the packaging cell line (CymR-TAD is encoded on a transgene plasmid and transfected), and a chemical-based induction system in the producer cell line, without changing the underlying configuration of the integrated AAV and Adenoviral genes (i.e. a packaging cell line is used to make the producer cell line). All of these processes can be used in a Good Manufacturing Process (GMP) with reduced safety concerns.

(30) The systems of the invention also provide a high level of control over AAV gene transcription, with reduced ‘off’ state expression levels and high cony state expression levels.

(31) In one embodiment, the invention provides a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, (ii) a cap gene, and (iii) a rep gene, wherein the rep gene preferably encodes Rep78 and Rep68, in the above 5′-3′ order, wherein the cap gene and the rep gene are both operably associated with the promoter.

(32) In another embodiment, the invention provides a cell line, comprising: (A) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, (ii) a cap gene, and (iii) a rep gene, wherein the rep gene preferably encodes Rep78 and Rep68, in the above 5′-3′ order, wherein the cap gene and the rep gene are both operably associated with the promoter; and optionally one of more of the following: (B) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter (preferably a minimal promoter) comprising one or more cumate operator sites, operably associated with (ii) a gene encoding a rc-CymR-TAD polypeptide; (C) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter (preferably a constitutive promoter), operably associated with (ii) a gene encoding CymR; (D) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter (preferably a constitutive promoter), operably associated with (ii) a gene encoding a CymR-TAD polypeptide; (E) a nucleic acid molecule, vector or plasmid comprising: (i) one or more promoters, which may or may not comprise one or more cumate operator sites, operably associated with (ii) one or more adenoviral genes which are competent to support AAV production, preferably E4 and/or VAI-RNA; (F) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter, operably associated with (ii) a transgene, wherein the promoter and transgene are flanked by ITRs.

(33) In a further embodiment, there is provided a process for producing an AAV cell line, the process comprising the steps of integrating into the genome of the cells of the cell line: (A) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, (ii) a cap gene, and (iii) a rep gene, wherein the rep gene preferably encodes Rep78 and Rep68, in the above 5′-3′ order, wherein the cap gene and the rep gene are both operably associated with the promoter, thereby producing a cell that inducibly expresses viral cap and rep genes; and optionally one of more of the following: (B) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter (preferably a minimal promoter) comprising one or more cumate operator sites, operably associated with (ii) a gene encoding a rc-CymR-TAD polypeptide; (C) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter (preferably a constitutive promoter), operably associated with (ii) a gene encoding CymR; (E) a nucleic acid molecule, vector or plasmid comprising: (i) one or more promoters, which may or may not comprise one or more cumate operator sites, operably associated with (ii) one or more adenoviral genes

which are competent to support AAV production, preferably E4 and/or VAI-RNA.
(34) The invention also provides a method of inducing transcription of cap and rep genes in a cell, the method comprising the steps: inducing the transcription of cap and rep genes in a cell which comprises in its genome: (A) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, (ii) a cap gene, and (iii) a rep gene, wherein the rep gene preferably encodes Rep78 and Rep68, in the above 5′-3′ order, wherein the cap gene and the rep gene are both operably associated with the promoter, wherein the inducing is by the presence, in the cell, of a CymR-TAD polypeptide.
(35) The invention also provides a method of inducing transcription of cap and rep genes in a cell, the method comprising the steps: inducing the transcription of cap and rep genes in a cell which comprises in its genome: (A) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, (ii) a cap gene, and (iii) a rep gene, wherein the rep gene preferably encodes Rep78 and Rep68, in the above 5′-3′ order, wherein the cap gene and the rep gene are both operably associated with the promoter; (B) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, operably associated with (ii) a gene encoding a rc-CymR-TAD polypeptide; (C) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter, operably associated with (ii) a gene encoding CymR; and optionally (D) a nucleic acid molecule, vector or plasmid comprising: (i) one or more promoters, which may or may not comprise one or more cumate operator sites, operably associated with (ii) one or more adenoviral genes which are competent to support AAV production, preferably E4 and/or VAI-RNA; and/or (E) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter, operably associated with (ii) a transgene, wherein the promoter and transgene are flanked by ITRs, wherein (A), (B), (C), (D) and (E), when present, are preferably all integrated into the genome of the cell, wherein the inducing is by the presence, in the cell, of cumate.
(36) The nucleic acid molecule may be DNA or RNA, preferably DNA. The nucleic acid molecule may be single- or double-stranded, preferably double-stranded.
(37) In one embodiment, the invention provides a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, (ii) a cap gene, and (iii) a rep gene, wherein the rep gene preferably encodes Rep78 and Rep68, in the above 5′-3′ order, wherein the cap gene and the rep gene are both operably associated with the promoter.
(38) In this embodiment, the promoter is preferably a minimal promoter. The cap gene and the rep gene are both operably associated with the promoter. This will result in a single primary transcript that comprises coding sequences from both cap and rep genes. As used herein, the term “minimal promoter” refers to any DNA sequence which is capable of recruiting the basal transcription complex to a DNA molecule. Preferably, the minimal promoter includes an InR box and/or a Downstream Promoter Element (DPE) box and/or a TATA box sequence. A minimal promoter excludes promoter elements which are normally found upstream of the TATA box, such as enhancer sequences. A minimal promoter is capable of initiating only low levels of transcription. The minimal promoter may also be called a basally inactive promoter (i.e. active only at a base level in the absence of other genetic elements).
(39) Preferably, the minimal promoter is a minimal pol II promoter. Examples of minimal promoters include promoters which are derived from the CMV, SV40, PGK (human or mouse), HSV TK, SFFV, Ubiquitin, Elongation Factor Alpha, CHEF-1, FerH, Grp78, RSV, Adenovirus E1A, CAG or CMV-Beta-Globin promoter.
(40) Preferably, the minimal promoter is derived from the cytomegalovirus immediate early (CMV) promoter, or a promoter of equal or increased strength compared to the CMV promoter in human cells and human cell lines (e.g. HEK-293 cells).
(41) In some embodiments, the promoter comprises one or more cumate operator sites. As used herein, the term cumate operator site (which may be abbreviated herein as “CuO”) refers to a DNA element which is capable of being bound by a cumate repressor protein (CymR). The CuO is therefore a DNA recognition sequence for CymR.
(42) Preferably, the CuO site has the sequence: AGAAACAAACCAACCTGTCTGTATTA (SEQ ID NO: 12) or a variant thereof which is capable of being bound by CymR.
(43) In some preferred embodiments, the promoter comprises a plurality of CuO sites, most preferably 6 CuO sites. Preferably, the sequence of the 6 CuO sites is:

AGAAACAAACCAACCTGTCTGTATTATCAAAGAAACAAACCAACCTGTCTGTATTAT
CAAAGAAACAAACCAACCTGTCTGTATTATCAAAGAAACAAACCAACCTGTCTGTAT
TATCAAAGAAACAAACCAACCTGTCTGTATTATCAAAGAAACAAACCAACCTGTCTG TATTA
(SEQ ID NO: 13) or a variant thereof which is capable of being bound by a plurality of CymR molecules.
(44) In some embodiments, the promoter comprises 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 CuO sites. In some embodiments, the promoter preferably comprises 4, 5, 6, 7, 8 or 9 CuO sites, most preferably 6 CuO sites.
(45) In other embodiments (e.g. the promoter which is operably associated with a gene encoding a rc-CymR-TAD polypeptide), the promoter comprises only 1, 2 or 3 CuO sites, preferably only 1 CuO site.
(46) Preferably, the CuO site(s) in the promoter are placed upstream of both the TATA box and the +1 site.
(47) The CuO site(s) may also be placed downstream of the +1 site at any distance up to the ATG start codon of the first coding sequence.
(48) In embodiments of the invention which comprise more than one CuO site, the CuO sites may be joined contiguously or non-contiguously. Preferably, the CuO sites are separated by linker nucleotide sequences. The linker nucleotide sequences may, for example, be 5-20, preferably 7-15 and most preferably about 10 nucleotides. In some embodiments, the linker sequences are 4 nucleotides.
(49) In some other embodiments (e.g. the promoter which is operably associated with a gene encoding a CymR or CymR-TAD polypeptide), the promoter is preferably a constitutive promoter.
(50) Examples of constitutive promoters include the CMV, SV40, PGK (human or mouse), HSV TK, SFFV, Ubiquitin, Elongation Factor Alpha, CHEF-1, FerH, Grp78, RSV, Adenovirus E1A, CAG or CMV-Beta-Globin promoter, or a promoter derived therefrom. Preferably, the promoter is the cytomegalovirus immediate early (CMV) promoter, or a promoter which is derived therefrom, or a promoter of equal or increased strength compared to the CMV promoter in human cells and human cell lines (e.g. HEK-293 cells). Preferably, the promoter which is operably-associated with the cap and rep genes is not an AAV promoter, e.g. it is not an AAV p5, p19 or p40 promoter.
(51) The nucleic acid molecule of the invention comprises a rep gene. The rep gene preferably encodes Rep78 and Rep68. As used herein, the term “rep gene” refers to a gene that encodes one or more open reading frames (ORFs), wherein each of said ORFs encodes an AAV Rep non-structural protein, or variant or derivative thereof. These AAV Rep non-structural proteins (or variants or derivatives thereof) are involved in AAV genome replication and/or AAV genome packaging.
(52) The structure of the wild-type AAV genome, illustrating the organisation of the wild-type rep and cap genes, is shown in FIG. **2**.
(53) The wild-type rep gene comprises three promoters: p5, p19 and p40. Two overlapping messenger ribonucleic acids (mRNAs) of different lengths can be produced from p5 and from p19. Each of these mRNAs contains an intron which can be either spliced out or not using a single splice donor site and two different splice acceptor sites. Thus, six different mRNAs can be formed, of which only four are functional. The two mRNAs that fail to remove the intron (one transcribed from p5 and one from p19) read through to a shared terminator sequence and encode Rep78 and Rep52, respectively. Removal of the intron and use of the 5′-most splice acceptor site does not result in production of any functional Rep protein—it cannot produce the correct Rep68 or Rep40 proteins as the frame of the remainder of the sequence is shifted, and it will also not produce the correct C-terminus of Rep78 or Rep52 because their terminator is spliced out. Conversely, removal of the intron and use of the 3′ splice acceptor will include the correct C-terminus for Rep68 and Rep40, whilst splicing out the terminator of Rep78 and Rep52. Hence the only functional splicing either avoids splicing out the intron altogether (producing Rep78 and Rep52) or uses the 3′ splice acceptor (to produce Rep68 and Rep40). Consequently, four different functional Rep proteins with overlapping sequences can be synthesized from these promoters.
(54) In the wild-type rep gene, the p40 promoter is located at the 3′ end. Transcription of the Cap proteins (VP1, VP2 and VP3) is initiated from this promoter in the wild-type AAV genome.
(55) The four wild-type Rep proteins are Rep78, Rep68, Rep52 and Rep40. Hence the wild-type rep gene is one which encodes the four Rep proteins Rep78, Rep68, Rep52 and Rep40.
(56) Rep78 and 68 can specifically bind the hairpin formed by the ITR and cleave it at a specific region (i.e. the terminal resolution site) within the hairpin. In the wild-type virus, they are also necessary for the AAV-specific integration of the AAV genome. Rep 78 and Rep68 are transcribed under control of the p5 promoter in the wild type virus, and the difference between them reflects removal (or not) of an intron by

splicing, hence they have different C terminal protein composition.
(57) Rep52 and Rep40 are involved in genome packaging. Rep52 and Rep40 are transcribed under control of the p19 promoter in the wild type virus, and the difference between them reflects removal (or not) of an intron by splicing, hence they have different C terminal protein composition.
(58) All four Rep proteins bind ATP and possess helicase activity. They up-regulate transcription from the p40 promoter, but down-regulate both p5 and p19 promoters.
(59) As used herein, the term “rep gene” includes wild-type rep genes and derivatives thereof; and artificial rep genes which have equivalent functions.
(60) In one embodiment, the rep gene encodes functional Rep78, Rep68, Rep52 and Rep40 proteins.
(61) In a preferred example of this embodiment, Rep78 and Rep 68 are translated by ribosomes docking 5′ to the Rep78 and Rep68 ATG start codon, thus allowing production of both of these proteins. In this example, the Rep78 and Rep68 open reading frames contain an active p40 promoter that provides the expression of both Rep52 and Rep40.
(62) In some embodiments of the invention, the function of one or more of the p5, p19 and p40 promoters is removed/disabled, for example by codon-changing and/or removal of the TATA box, in order to prevent unwanted initiation of transcription from that promoter.
(63) Preferably, the p5 promoter is non-functional (i.e. it cannot be used to initiate transcription). More preferably, the p5 promoter is replaced with the IRES (thus removing the function of the p5 promoter). This allows Rep78 or Rep68 to be transcribed in the same mRNA as the cap genes, but translation of the Rep78 and Rep68 proteins will be under the control of the IRES.
(64) A further advantage of the removal of the p5 promoter is that, in the wild-type virus, the p5 promoter is bound by and is activated by the E2A DNA-binding protein (DBP). Hence the removal of the p5 promoter means that the E2A gene is not required (e.g. in a Helper Plasmid) to produce virus particles.
(65) In one embodiment, the rep gene does not have a p5 promoter upstream. In another embodiment, the p5 promoter is not used in AAV packaging.
(66) Preferably, the p19 promoter within the rep gene is functional.
(67) In some embodiments, the function of the p40 promoter is removed/disabled within the Rep gene by one or more codon changes.
(68) The cap gene is preferably relocated and its transcription is placed under control of an alternative promoter (e.g. CMV immediate early promoter).
(69) There is a degree of redundancy between the function of the different Rep proteins and hence, in some embodiments of the invention, not all of the Rep proteins are required.
(70) In some embodiments, the rep gene only encodes one, two, three or four of Rep78, Rep68, Rep52 and Rep40, preferably one, two or four of Rep78, Rep68, Rep52 and Rep40.
(71) In some embodiments, the rep gene does not encode one or more of Rep78, Rep68, Rep52 and Rep40.
(72) In some embodiments, the rep gene encodes Rep78 and Rep52, but does not encode Rep68 or Rep40. In this embodiment, the splice donor site remains in the DNA but both the 5′ and 3′ splice acceptor sites are removed. Hence the intron cannot be removed by splicing and transcription continues through to the terminator sequence for Rep78 and Rep52 (which is common to both). The Rep78 protein is transcribed in the same mRNA as the cap gene (hence is driven by the same promoter), and translation of Rep78 is driven by the IRES. Transcription of Rep52 is driven by the p19 promoter; hence it forms a separate mRNA and is translated by 5′ m.sup.7G cap-dependent docking at the ribosome. Accordingly, Rep68 and Rep40 cannot be produced in this embodiment.
(73) In other embodiments, the rep gene encodes Rep68 and Rep40, but does not encode Rep78 or Rep52. In this embodiment, the intronic sequence between the splice donor and 3′ splice acceptor is removed at the DNA level, placing the C terminus of Rep68 and Rep40 in frame with the upstream coding sequence. Hence Rep68 and Rep40 (but not Rep78 and Rep52) are produced. For clarity, Rep68 is transcribed in the same mRNA as the Cap proteins and it is translated under control of the IRES. In contrast, Rep40 is transcribed into a separate mRNA by the p19 promoter and it is translated by 5′ m.sup.7G cap docking at the ribosome.
(74) In some preferred embodiments, the rep gene encodes Rep78 and Rep68, but does not encode Rep52 or Rep40. This may be achieved by mutating the p19 promoter (e.g. inserting a mutation at the p19 TATA

box).
(75) In some embodiments, the rep gene encodes Rep52 and Rep40, but does not encode Rep78 or Rep68. This may be achieved by including just the coding sequence from the ATG of Rep52/40.
(76) Preferably, the rep gene encodes Rep78 and Rep68, but does not encode Rep52 or Rep40.
(77) As used above, the term “encodes” means that the rep gene encodes a functional form of that Rep protein. Similarly, the term “does not encode” means that the rep gene does not encode a functional form of that Rep protein.
(78) In the absence of sufficient Rep proteins, lower titres (e.g. genome copies) would be observed (which could be determined by qPCR), due to the fact that there is less ITR plasmid to be packaged and that it would not be effectively packaged. The observation might also include an exaggerated empty:full particle ratio; this could be determined by ELISA or optical density measurement.
(79) The wild-type AAV (serotype 2) rep gene nucleotide sequence is given in SEQ ID NO: 1. The wild-type AAV (serotype 2) Rep78, Rep68, Rep52 and Rep40 amino acid sequences are given in SEQ ID NOs: 2, 3, 4 and 5, respectively. The wild-type AAV (serotype 2) nucleotide sequence encoding Rep78 is given in SEQ ID NO: 6. The wild-type AAV (serotype 2) nucleotide sequence encoding Rep68 is given in SEQ ID NO: 7.
(80) The wild-type AAV (serotype 2) nucleotide sequence encoding Rep52 is given in SEQ ID NO: 8. The wild-type AAV (serotype 2) nucleotide sequence encoding Rep 40 is given in SEQ ID NO: 9.
(81) In one embodiment, the term “rep gene” refers to a nucleotide sequence having at least 70%, 80%, 85%, 90%, 95%, 99% or 100% sequence identity to SEQ ID NO: 1 and which encodes one or more Rep78, Rep68, Rep52 and Rep40 polypeptides.
(82) In another embodiment, the term “rep gene” refers to a nucleotide sequence having at least 70%, 80%, 85%, 90%, 95%, 99% or 100% sequence identity to SEQ ID NO: 6 and which encodes functional Rep78 and/or Rep52 polypeptides (and preferably does not encode functional Rep68 or Rep40 polypeptides).
(83) In another embodiment, the term “rep gene” refers to a nucleotide sequence having at least 70%, 80%, 85%, 90%, 95%, 99% or 100% sequence identity to SEQ ID NO: 7 and which encodes functional Rep68 and/or Rep40 polypeptides (and preferably does not encode functional Rep78 or Rep52 polypeptides).
(84) In another embodiment, the term “rep gene” refers to a nucleotide sequence having at least 70%, 80%, 85%, 90%, 95%, 99% or 100% sequence identity to SEQ ID NO: 8 and which encodes a functional Rep52 polypeptide (and preferably does not encode a functional Rep78 polypeptide).
(85) In another embodiment, the term “rep gene” refers to a nucleotide sequence having at least 70%, 80%, 85%, 90%, 95%, 99% or 100% sequence identity to SEQ ID NO: 9 and which encodes a functional Rep40 polypeptide (and preferably does not encode functional Rep68 polypeptide).
(86) In another embodiment, the term “rep gene” refers to a nucleotide sequence having at least 90%, 95%, 99% or 100% sequence identity to a nucleotide sequence which encodes SEQ ID NO: 2 and which encodes functional Rep78 and/or Rep52 polypeptides (and preferably does not encode functional Rep68 or Rep40 polypeptides).
(87) In another embodiment, the term “rep gene” refers to a nucleotide sequence having at least 90%, 95%, 99% or 100% sequence identity to a nucleotide sequence which encodes SEQ ID NO: 3 and which encodes functional Rep68 and/or Rep40 polypeptides (and preferably does not encode functional Rep78 or Rep52 polypeptides).
(88) In another embodiment, the term “rep gene” refers to a nucleotide sequence having at least 90%, 95%, 99% or 100% sequence identity to a nucleotide sequence which encodes SEQ ID NO: 4 and which encodes a functional Rep52 polypeptide (and preferably does not encode a functional Rep78 polypeptide).
(89) In another embodiment, the term “rep gene” refers to a nucleotide sequence having at least 90%, 95%, 99% or 100% sequence identity to a nucleotide sequence which encodes SEQ ID NO: 5 and which encodes a functional Rep40 polypeptide (and preferably does not encode functional Rep68 polypeptide).
(90) In some embodiments, the nucleic acid molecule of the invention does not encode a functional Rep78 polypeptide. In some embodiments, the nucleic acid molecule of the invention does not encode a functional Rep68 polypeptide. In some embodiments, the nucleic acid molecule of the invention does not

encode a functional Rep52 polypeptide. In some embodiments, the nucleic acid molecule of the invention does not encode a functional Rep40 polypeptide.
(91) The nucleic acid molecule also comprises a cap gene. As used herein, the term "cap gene" refers to a gene that encodes one or more open reading frames (ORFs), wherein each of said ORFs encodes an AAV Cap structural protein, or variant or derivative thereof. These AAV Cap structural proteins (or variants or derivatives thereof) form the AAV capsid.
(92) The three Cap proteins must function to enable the production of an infectious AAV virus particle which is capable of infecting a suitable cell. The three Cap proteins are VP1, VP2 and VP3, which are generally 87 kDa, 72 kDa and 62 kDa in size, respectively. Hence the cap gene is one which encodes the three Cap proteins VP1, VP2 and VP3.
(93) In the wild-type AAV, these three proteins are translated from the p40 promoter to form a single mRNA. After this mRNA is synthesized, either a long or a short intron can be excised, resulting in the formation of a 2.3 kb or a 2.6 kb mRNA.
(94) Usually, especially in the presence of adenovirus, the long intron is excised. In this form the first AUG codon, from which the synthesis of VP1 protein starts, is cut out, resulting in a reduced overall level of VP1 protein synthesis. The first AUG codon that remains is the initiation codon for VP3 protein. However, upstream of that codon in the same open reading frame lies an ACG sequence (encoding threonine) which is surrounded by an optimal Kozak context. This contributes to a low level of synthesis of VP2 protein, which is actually VP3 protein with additional N terminal residues, as is VP1.
(95) If the long intron is spliced out, and since in the major splice the ACG codon is a much weaker translation initiation signal, the ratio at which the AAV structural proteins are synthesized in vivo is about 1:1:10, which is the same as in the mature virus particle. The unique fragment at the N-terminus of VP1 protein has been shown to possess phospholipase A2 (PLA2) activity, which is probably required for the releasing of AAV particles from late endosomes.
(96) The AAV capsid is composed of 60 capsid protein subunits (VP1, VP2, and VP3) that are arranged in an icosahedral symmetry in a ratio of 1:1:10, with an estimated size of 3.9 MDa.
(97) As used herein, the term "cap gene" includes wild-type cap genes and derivatives thereof, and artificial cap genes which have equivalent functions.
(98) The AAV (serotype 2) cap gene nucleotide sequence and Cap polypeptide sequences are given in SEQ ID NOs: 10 and 11, respectively.
(99) As used herein, the term "cap gene" refers preferably to a nucleotide sequence having the sequence given in SEQ ID NO: 10 or a nucleotide sequence encoding SEQ ID NO: 11; or a nucleotide sequence having at least 70%, 80%, 85% 90%, 95% or 99% sequence identity to SEQ ID NO: 10 or at least 80%, 90%, 95% or 99% nucleotide sequence identity to a nucleotide sequence encoding SEQ ID NO: 11, and which encodes VP1, VP2 and VP3 polypeptides.
(100) The rep and cap genes are preferably viral genes or derived from viral genes. More preferably, they are AAV genes or derived from AAV genes. In some embodiments, the AAV is an Adeno-associated dependoparvovirus A. In other embodiments, the AAV is an Adeno-associated dependoparvovirus B.
(101) 11 different AAV serotypes are known. All of the known serotypes can infect cells from multiple diverse tissue types. Tissue specificity is determined by the capsid serotype. The AAV may be from serotype 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or 11. Preferably, the AAV is serotype 1, 2, 5, 6, 7, 8 or 9. Most preferably, the AAV serotype is 5 (i.e. AAVS).
(102) The rep and cap genes (and each of the protein-encoding ORFs therein) may be from one or more different viruses (e.g. 2, 3 or 4 different viruses). For example, the rep gene may be from AAV2, whilst the cap gene may be from AAVS.
(103) It is recognised by those in the art that the rep and cap genes of AAV vary by clade and isolate. The sequences of these genes from all such clades and isolates are encompassed herein, as well as derivatives thereof.
(104) The cap gene and rep gene are present in the nucleic acid in this cap-rep 5′.fwdarw.3′ order. However, since Rep52 and/or Rep40 may be transcribed from their own p19 promoter, the position of the coding sequence which encodes Rep52 and/or Rep40 may be varied. For example, the coding sequence which encodes Rep52 and/or Rep40 may be placed upstream or downstream of the cap genes and rep genes which encode Rep78/68; or indeed on the reverse strand of the nucleic acid of the invention or on a

different nucleic acid.
(105) Translation of the cap gene is preferably initiated from the standard 5′ m.sup.7G-cap at the 5′ end of the mRNA.
(106) A single mRNA transcript will be produced which encodes both the Cap and Rep polypeptides. Additional measures may be taken in order to promote appropriate levels of translation of the (downstream) Rep polypeptide. These measures include: (a) inserting an IRES site between the cap and rep genes; (b) production of a Cap-Rep fusion polypeptide and its subsequent cleavage into separate polypeptides; and (c) use of alternative splicing to produce separate Cap and Rep polypeptides. Each of these measures are discussed further below.
(107) The rep gene may be operably-associated with an Internal Ribosome Entry Site (IRES). The IRES regulates the translation of the rep mRNA. IRESs are distinct regions of nucleic acid molecules that are able to recruit eukaryotic ribosomes to the mRNA in a process which is known as cap-independent translation. IRESs are commonly located in the 5′-UTRs of RNA viruses. They facilitate translation of the viral RNAs in a cap-independent manner.
(108) Examples of viral IRESs include Picornavirus IRES (Encephalomyocarditis virus, EMCV IRES), Aphthovirus IRES (Foot-and-mouth disease virus, FMDV IRES), Kaposi's sarcoma-associated herpes virus IRES, Hepatitis A IRES, Hepatitis C IRES, Pestivirus IRES, Cripavirus internal ribosome entry site (IRES), *Rhopalosiphum padi* virus internal ribosome entry site (IRES) and 5′-Leader IRES and intercistronic IRES in the 1.8-kb family of immediate early transcripts (IRES)1. The invention also encompasses non-natural derivatives of the above IRESs which retain the capacity to recruit eukaryotic ribosomes to the mRNA.
(109) In some preferred embodiments, the IRES is an encephalomyocarditis virus (EMCV) IRES. In one embodiment of the invention, the nucleotide sequence of the EMCV IRES is as given in SEQ ID NO: 14 or a nucleotide sequence having at least 80%, more preferably at least 85%, 90% or 95% sequence identity thereto and which encodes an IRES.
(110) In other embodiments, the IRES is a Foot-and-mouth disease virus (FMDV) IRES. In one embodiment of the invention, the nucleotide sequence of the FMDV IRES is as given in SEQ ID NO: 15 or a nucleotide sequence having at least 80%, more preferably at least 85%, 90% or 95% sequence identity thereto and which encodes an IRES.
(111) The rep gene is operably-associated with the IRES. Preferably, the IRES is located downstream of the cap gene and upstream of the translation start site for Rep 78/68. In some embodiments, the IRES replaces the wild-type p5 promoter. A further advantage of the removal of the p5 promoter is that, in the wild-type virus, the p5 promoter is bound by and is activated by the E2A DNA-binding protein (DBP). Hence the removal of the p5 promoter means that the E2A gene is not required (e.g. in a Helper Plasmid) to produce virus particles.
(112) In other embodiments, the Cap and Rep polypeptides are produced as fusion polypeptide. In this case, the stop codon at the end of the cap gene is replaced with one or more codons which encode linker amino acids, and it is ensured that the coding sequences of the cap and rep genes are in frame. A single polypeptide which comprises the amino acid sequences of both Cap and Rep polypeptides is therefore produced as a fusion polypeptide. If an appropriate cleavable amino acid linker sequence is used, then the fusion polypeptide may be cleaved into separate Cap and Rep polypeptides.
(113) In other embodiments, the Cap and Rep polypeptides are produced by alternative splicing. In this case the cap gene(s) would contain a splice donor sequence upstream of the first cap gene ATG start codon but downstream of the promoter +1 position driving transcription of the cap gene(s). After the stop codon of the cap gene(s) a splice acceptor site would be added to allow splicing over the cap gene from the 5′ donor splice site to the 3′ acceptor splice site. After the splice acceptor site (i.e. downstream or 3′), the rep gene coding sequence(s) would be placed to allow expression from spliced mRNA transcripts. In the interest of clarity, in this model the cap gene(s) are placed within an intron that is spliced when rep gene expression is required.
(114) Sequences known to mediate splicing are known to those in the art, and sequences would have to be selected based on their strength and propensity to induce splicing to ensure that an optimal level of cap gene(s) and rep genes were expressed to enable efficient viral particle production. Ensuring the optimal relative stoichiometry of the rep and cap mRNA encoding transcripts is therefore important.

(115) In other embodiments, Cap and Rep polypeptides may be produced by encoding a 2A peptide between the main polypeptides. This 2A sequence could derive from porcine teschovirus-1 (P2A), *Thosea asigna* virus 2A (T2A), FMDV 2A (abbreviated herein as F2A) or equine rhinitis A virus (ERAV) 2A (E2A). When a polypeptide is produced that contains a 2A sequence, the polypeptide is cleaved into two polypeptides by the 2A sequence. As such, the two remaining proteins are at equal stoichiometries. This is undesirable in the case of AAV Rep and AAV Cap where excess expression of Cap is deemed to be advantageous. Therefore, although one embodiment of the invention may use a 2A sequence, it is not the preferred method.
(116) The production of stable cell lines in mammalian culture typically requires a method of selection to promote the growth of cells containing any exogenously-added DNA. Preferably, one or more of the nucleic acid molecules of the invention additionally comprise a selection gene or an antibiotic resistance gene. To this end, a range of genes are known that provide resistance to specific compounds when the DNA encoding them is inserted into a mammalian cell genome. Preferably, the selection gene is puromycin N-acetyl-transferase (Puro), hygromycin phosphotransferase (Hygro), blasticidin s deaminase (Blast), Neomycin phosphotransferase (Neo), glutathione S-transferase (GS), zeocin resistance gene (Sh ble) or dihydrofolate reductase (DHFR). Each of these genes provides resistance to a small molecule known to be toxic to mammalian cells, or in the case of GS provides a method for cells to generate glutathione in the absence of glutathione in the growth media.
(117) In a preferred embodiment of the invention, the resistance gene is Puro. This gene is particularly effective because many of the cell lines used in common tissue culture are not resistant to Puro; this cannot be said for Neo where many, particularly HEK 293 derivatives, are already Neo resistant due to previous genetic manipulations by researchers (e.g. HEK 293T cells). Puro selection also has the advantage of being toxic over a short time window (<72 hours), and hence it allows variables to be tested rapidly and cells that do not harbour the exogenous DNA to be inserted into the genome are rapidly removed from the culture systems. This cannot be said of some other selection methods such as Hygro, where toxicity is much slower onset.
(118) The development of stable cell lines using selection genes (e.g. Puro) requires that the resistance gene must be expressed in the cells. This can be achieved through a variety of methods including, but not limited to, internal ribosome entry sites (IRES), 2A cleavage systems, alternative splicing, and dedicated promoters.
(119) In a preferred embodiment of the invention, the selection gene will be expressed from a dedicated promoter. This promoter will preferably transcribe in human cells at lower levels than the dedicated promoters driving the rep or cap genes.
(120) Each of the genes in the nucleic acid molecules which encode a polypeptide or RNA will preferably be operably-associated with one or more regulatory elements. This ensures that the polypeptide or RNA is expressed at the desired level and at the desired time. In this context, the term “regulatory elements” includes one or more of an enhancer, promoter, intron, polyA, insulator or terminator.
(121) The genes used in the AAV plasmids or vectors disclosed herein are preferably separated by polyA signals and/or insulators in an effort to keep transcriptional read-through to other genes to a minimum.
(122) While some advantages may be obtained by using copies of the same regulatory element (e.g. promoter sequence) with more than one polypeptide or RNA-encoding nucleotide sequence (in terms of their co-ordinated expression), in the context of this invention, it is highly desirable to use different regulatory elements with each polypeptide or RNA-encoding nucleotide sequence. In this way, the risk of homologous recombination between these regulatory elements is reduced.
(123) In some embodiments, the nucleic acid molecule of the invention (or the vector or plasmid comprising it or cell) additionally comprises one or more genes encoding one or more Adenovirus E4 polypeptides. Preferably, the nucleic acid molecule (or the vector or plasmid comprising it or cell) additionally comprises an Adenovirus transcription unit which is required for AAV replication.
(124) The promoter which is operably associated with the E4 gene(s) may be inducible, repressible or constitutive. In some preferred embodiments, the promoter which is operably associated with the E4 gene(s) is a minimal promoter which comprises one or more cumate operator sites.
(125) Preferably, the promoter is activatable by the binding of the same CymR-TAD or reCymR-TAD which is capable of activating the promoter which is operably associated with the cap and rep genes. Most

preferably, the promoter is the same promoter as that which is operably associated with the cap and rep genes.
(126) The nucleic acid molecule of the invention will, most embodiments, be a plasmid or vector which is useful in the production of AAVs. In most embodiments, therefore, the nucleic acid molecule of the invention (or the vector or plasmid comprising it) will not comprise inverted terminal repeats (ITRs), except for the nucleic acid molecule comprising a transgene. In some embodiments, the nucleic acid molecule of the invention (or the vector or plasmid comprising it) will not comprise one or more genes selected from Adenovirus E1A, E1B, E4, E2a or VA.
(127) In some preferred embodiments, the nucleic acid molecule of the invention (or the vector or plasmid or plasmid system or cell comprising it) does not comprise the Adenovirus E2A gene.
(128) As used herein, the term “E2A” or “E2A gene” refers to a viral E2A gene or a variant or derivative thereof. Preferably, the E2A gene is from or derived from a human adenovirus, e.g. Ad5.
(129) In one embodiment of the invention, the nucleotide sequence of the Adenovirus E2A gene is as given in SEQ ID NO: 16 or a nucleotide sequence having at least 80%, more preferably at least 85%, 90% or 95% sequence identity thereto and which encodes a DNA-binding protein which aids elongation of viral DNA replication.
(130) In one embodiment of the invention, the nucleotide sequence for the AAV P5 promoter is absent.
(131) In another embodiment, there is provided a plasmid or vector comprising a nucleic acid molecule of the invention.
(132) Examples of preferred embodiments of the invention include nucleic acid molecules comprising the following elements in this order:
(133) 6×CuO+CMV minimal promoter-AAV2 cap gene-FMDV IRES-rep gene
(134) 6×CuO+p565 minimal promoter-AAV2 cap gene-EMCV IRES-rep gene
(135) 6×CuO+CMV minimal promoter-AAV2 cap gene-EMCV RES-rep gene
(136) Preferably, the term “rep gene” refers to a gene which encodes Rep78 and Rep68 polypeptides (but preferably does not encode functional Rep52 or Rep40 polypeptides).
(137) In some embodiments, the invention involves a nucleic acid molecule comprising: (i) a promoter, operably associated with (ii) a gene encoding a CymR polypeptide. In this molecule, the promoter is preferably a constitutive promoter. In some embodiments, this nucleic acid molecule is integrated into the genome of a host cell or cell line.
(138) In some embodiments, the invention involves a nucleic acid molecule, vector or plasmid comprising: (i) a promoter, operably associated with (ii) a gene encoding a CymR polypeptide. In this molecule, the promoter is preferably a constitutive promoter. This nucleic acid may be used to induce expression of Rep and Cap polypeptides from the CuO-cap-rep nucleic acid molecule of the invention. Preferably, this nucleic acid molecule is not integrated into the genome of a host cell or cell line.
(139) In some embodiments, the invention involves a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, operably associated with (ii) a gene encoding a rc-CymR-TAD polypeptide. In this molecule, the promoter is preferably a minimal promoter (as defined herein). In some embodiments, this nucleic acid molecule is integrated into the genome of a host cell or cell line. The promoter preferably comprises 1, 2 or 3, most preferably only one CuO site.
(140) The term “rc-CymR” refers to a reverse cumate-repressor protein. This is a mutant CymR polypeptide which binds CuO sites in the presence of cumate, but not in the absence of cumate. Such rcCymR mutants are known in the art. Compared to the wild-type CymR polypeptide sequence, one such mutant has the mutations Ala125Val, Glu142Gly and Met144Ile.
(141) Preferably, the CymR and rcCymR polypeptides used in the context of the current invention have a deletion of the wild-type amino acids Met Val Ile at the N-terminal. The preferred CymR- and rcCymR-encoding nucleotide sequences also have corresponding deletions. Preferred examples of the CymR and rcCymR nucleotide and polypeptide sequences are given in SEQ ID NOs: 17-20.
(142) The term “TAD” refers to a “transactivator domain”. Transactivator domains act by recruiting elements of the pre-initiation complex, for example TFIID, to the TATA box.
(143) TFIID is a composition of subunits including the TATA-binding protein (TBP) and TBP-associated factors (TAFs). TBP binding to the TATA-box is enough to initiate transcription in TATA-box containing

promoters. Once TFIID is bound to the promoter, RNA polymerase II and the remainder of the pre-initiation complex is recruited, thus allowing initiation of transcription.

(144) The TAD may therefore be a polypeptide or domain thereof which binds to at least one element of the transcription pre-initiation complex, e.g. to TFIID, to a TATA-binding protein (TBP) or to TBP-associated factor (TAF).

(145) In the context of this invention, the minimal promoter contains the TATA-box and downstream promoter elements, but all upstream promoter elements and enhancer sequences have been removed. They are replaced with DNA-binding sites for a DNA-binding protein, i.e. CuO.

(146) A transactivation domain (TAD) is a transcription factor scaffold which can bind either directly or indirectly with transcriptional co-activators (such as TAFs). By fusing these TADs to a DNA-binding protein (i.e. rcCymR), which itself is directed to the enhancer region (e.g. CuO) of a promoter by positioning of the DNA binding sites upstream of the TATA box, it is possible to target this transactivation activity to a promoter.

(147) The transactivation domain (TAD) is preferably a polypeptide or domain or part thereof, e.g. a domain of a protein which has the desired activity. Examples of TADs include VP16 (and concatamers termed VP32, VP64), p53, rTa, p65 and combinations thereof. The TAD may comprise a plurality of TADs, which may be the same (e.g. VP16-VP16) or different. Preferably, the TAD is from VP16 or p53. Both of these domains directly contact key TAFs to recruit TFIID. These TADs are derived from VP16 or p53 proteins (i.e. they are part of the larger protein).

(148) VP16 is a herpes simplex virus protein that normally induces immediate early gene transcription in the HSV. It can be multiplexed to form a stronger TAD.

(149) p53 is a tumour suppressor protein which contains several domains for DNA binding, tetramerization, and transcriptional activation. The TAD domains (AD1 and AD2, which can be used separately or together) are found at the N-terminal of the p53 monomer.

(150) The rcCymR-TAD may be fusion protein or the rcCymR and TAD may be joined by a linker (which may be an amino acid linker or non-amino acid linker, preferably an amino acid linker). They may also be joined, preferably at the C-terminus of the DNA targeting domain.

(151) There are many established algorithms available to align two amino acid or nucleic acid sequences. Typically, one sequence acts as a reference sequence, to which test sequences may be compared. The sequence comparison algorithm calculates the percentage sequence identity for the test sequence(s) relative to the reference sequence, based on the designated program parameters. Alignment of amino acid or nucleic acid sequences for comparison may be conducted, for example, by computer-implemented algorithms (e.g. GAP, BESTFIT, FASTA or TFASTA), or BLAST and BLAST 2.0 algorithms.

(152) Percentage amino acid sequence identities and nucleotide sequence identities may be obtained using the BLAST methods of alignment (Altschul et al. (1997), “Gapped BLAST and PSI-BLAST: a new generation of protein database search programs”, Nucleic Acids Res. 25:3389-3402; and http://www.ncbi.nlm.nih.gov/BLAST). Preferably the standard or default alignment parameters are used.

(153) Standard protein-protein BLAST (blastp) may be used for finding similar sequences in protein databases. Like other BLAST programs, blastp is designed to find local regions of similarity. When sequence similarity spans the whole sequence, blastp will also report a global alignment, which is the preferred result for protein identification purposes. Preferably the standard or default alignment parameters are used. In some instances, the “low complexity filter” may be taken off.

(154) BLAST protein searches may also be performed with the BLASTX program, score=50, wordlength=3. To obtain gapped alignments for comparison purposes, Gapped BLAST (in BLAST 2.0) can be utilized as described in Altschul et al. (1997) Nucleic Acids Res. 25: 3389. Alternatively, PSI-BLAST (in BLAST 2.0) can be used to perform an iterated search that detects distant relationships between molecules. (See Altschul et al. (1997) supra). When utilizing BLAST, Gapped BLAST, PSI-BLAST, the default parameters of the respective programs may be used.

(155) With regard to nucleotide sequence comparisons, MEGABLAST, discontiguous-megablast, and blastn may be used to accomplish this goal. Preferably the standard or default alignment parameters are used. MEGABLAST is specifically designed to efficiently find long alignments between very similar sequences. Discontiguous MEGABLAST may be used to find nucleotide sequences which are similar, but not identical, to the nucleic acids of the invention.

(156) The BLAST nucleotide algorithm finds similar sequences by breaking the query into short subsequences called words. The program identifies the exact matches to the query words first (word hits). The BLAST program then extends these word hits in multiple steps to generate the final gapped alignments. In some embodiments, the BLAST nucleotide searches can be performed with the BLASTN program, score=100, wordlength=12.
(157) One of the important parameters governing the sensitivity of BLAST searches is the word size. The most important reason that blastn is more sensitive than MEGABLAST is that it uses a shorter default word size (11). Because of this, blastn is better than MEGABLAST at finding alignments to related nucleotide sequences from other organisms. The word size is adjustable in blastn and can be reduced from the default value to a minimum of 7 to increase search sensitivity.
(158) A more sensitive search can be achieved by using the newly-introduced discontiguous megablast page (www.ncbi.nlm.nih.gov/Web/Newsltr/FallWinter02/blastlab.html). This page uses an algorithm which is similar to that reported by Ma et al. (Bioinformatics. 2002 March; 18(3): 440-5). Rather than requiring exact word matches as seeds for alignment extension, discontiguous megablast uses non-contiguous word within a longer window of template. In coding mode, the third base wobbling is taken into consideration by focusing on finding matches at the first and second codon positions while ignoring the mismatches in the third position. Searching in discontiguous MEGABLAST using the same word size is more sensitive and efficient than standard blastn using the same word size. Parameters unique for discontiguous megablast are: word size: 11 or 12; template: 16, 18, or 21; template type: coding (0), non-coding (1), or both (2).
(159) In some embodiments, the BLASTP 2.5.0+ algorithm may be used (such as that available from the NCBI) using the default parameters.
(160) In other embodiments, a BLAST Global Alignment program may be used (such as that available from the NCBI) using a Needleman-Wunsch alignment of two protein sequences with the gap costs: Existence 11 and Extension 1.
(161) One method for the production of recombinant AAVs is based on the transfection of some or all elements that are required for AAV production into host cells, such as HEK293 cells, or engineered packaging cell lines as described elsewhere.
(162) For HEK293 host cells, this generally involves co-transfection with 3 vectors or plasmids: (a) an AAV ITR-containing plasmid, carrying the gene (i.e. transgene) of interest; (b) a plasmid that carries the AAV rep-cap genes; and (c) a plasmid that provides the necessary helper genes isolated from adenovirus.
(163) The invention therefore provides a kit comprising: (A) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, (ii) a cap gene, and (iii) a rep gene, wherein the rep gene preferably encodes Rep78 and Rep68, in the above 5′-3′ order, wherein the cap gene and the rep gene are both operably associated with the promoter; and optionally one of more of the following: (B) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter (preferably a minimal promoter) comprising one or more cumate operator sites, operably associated with (ii) a gene encoding a rc-CymR-TAD polypeptide; (C) a nucleic acid molecule, vector or plasmid comprising (i) a promoter (preferably a constitutive promoter), operably associated with (ii) a gene encoding a CymR; (D) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter (preferably a constitutive promoter), operably associated with (ii) a gene encoding a CymR-TAD polypeptide; (E) a nucleic acid molecule, vector or plasmid comprising: (i) one or more promoters, which may or may not comprise one or more cumate operator sites, operably associated with (ii) one or more adenoviral genes which are competent to support AAV production, preferably E4 and/or VAI-RNA; (F) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter, operably associated with (ii) a transgene, wherein the promoter and transgene are flanked by ITRs.
(164) The nucleic acid molecules may, for example, be plasmids or vectors.
(165) In some embodiments, the kit comprises nucleic acid molecules (A), (D) and (F). In other embodiments, the kit comprises nucleic acid molecules (A), (B), (C) and (F). Nucleic acid molecule (E) may optionally be added to the above embodiments.
(166) The kit may also comprise cumate.
(167) In some embodiments, the selected nucleic acid molecules are present on the same vector or plasmid. This therefore reduces the number of separate vectors or plasmids to be used in the transfection

process. Preferably, nucleic acid molecules (D) and (F) are present on the same nucleic acid molecule.
(168) The kit may also additionally comprise a Helper Plasmid comprising one or more viral genes selected from adenovirus E1A, E1B, E4 and VA.
(169) In some embodiments of the invention, the Helper Plasmid comprises an E2A gene. In other embodiments, the Helper Plasmid does not comprise an E2A gene. In the latter case, the omission of the E2A gene reduces considerably the amount of DNA which is needed in the Helper Plasmid.
(170) The kit may also additionally comprise a mammalian host cell (e.g. HEK293), optionally comprising one or more viral genes selected from E1A, E1B, E4 and VA expressible from the host cell genome. In some embodiments of the invention, the mammalian host cell additionally comprises an E2A gene expressible from the host cell genome. In other embodiments, the mammalian host cell does not comprise an E2A gene.
(171) In some embodiments, one or more of the nucleic acids may be delivered to the cell or cell lines using an Adenoviral vector. This will encode Adenoviral genes but may also encode the CymR-TAD and/or transgene. This may have advantages due to the efficiency of gene delivery to AAV-producing cells that can be achieved by an Adenoviral vector.
(172) The kit may also contain reagents and/or materials for purification of the AAV particles such as those involved in the density banding and purification of viral particles, e.g. one or more of centrifuge tubes, Iodixanol, dialysis buffers and dialysis cassettes.
(173) In another embodiment, the invention provides a cell line, comprising: (A) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, (ii) a cap gene, and (iii) a rep gene, wherein the rep gene preferably encodes Rep78 and Rep68, in the above 5′-3′ order, wherein the cap gene and the rep gene are both operably associated with the promoter; and optionally one of more of the following: (B) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter (preferably a minimal promoter) comprising one or more cumate operator sites, operably associated with (ii) a gene encoding a rc-CymR-TAD polypeptide; (C) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter (preferably a constitutive promoter), operably associated with (ii) a gene encoding a CymR; (D) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter (preferably a constitutive promoter), operably associated with (ii) a gene encoding a CymR-TAD polypeptide; (E) a nucleic acid molecule, vector or plasmid comprising: (i) one or more promoters, which may or may not comprise one or more cumate operator sites, operably associated with (ii) one or more adenoviral genes which are competent to support AAV production, preferably E4 and/or VAI-RNA; (F) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter, operably associated with (ii) a transgene, wherein the promoter and transgene are flanked by ITRs.
(174) The nucleic acid molecules may, for example, be plasmids or vectors.
(175) In some embodiments, the cell line is a packaging cell line. In other embodiments, the cell line is a producer cell line.
(176) In some preferred embodiments, the packaging cells comprise nucleic acid molecules, vectors or plasmids (A) and (E). In other preferred embodiments, the packaging cells comprise nucleic acid molecules, vectors or plasmids (A), (B), (C) and (E). In other preferred embodiments, the producer cells comprise nucleic acid molecules, vectors or plasmids (A), (B), (C), (E) and (F).
(177) In some preferred embodiments, each of (A), (B), (C), (E), and (F), when present, are integrated into the genome of the cell line, wherein the specified genes are expressed or expressible therefrom.
(178) The cells may be isolated cells, e.g. they are not situated in a living animal or mammal. Examples of mammalian cells include those from any organ or tissue from humans, mice, rats, hamsters, monkeys, rabbits, donkeys, horses, sheep, cows and apes. Preferably, the cells are human cells. The cells may be primary or immortalised cells.
(179) Preferred cells include HEK-293 or a derivative thereof (e.g. HEK 293T, HEK-293E, HEK-293 FT, HEK-293S, HEK-293SG, HEK-293 FTM, HEK-293SGGD, HEK-293A), MDCK, C127, A549, HeLa, CHO, mouse myeloma, PerC6, 911 and Vero cell lines. HEK-293 cells have been modified to contain the E1A and E1B proteins and this obviates the need for these proteins to be supplied on a Helper Plasmid. Similarly, PerC6 and 911 cells contain a similar modification and can also be used. Most preferably, the human cells are HEK293, HEK293T, HEK293A, PerC6 or 911. Other preferred cells include CHO and VERO cells.

(180) Preferably, the cells of the invention are capable of inducibly-expressing the rep and cap genes and/or inducibly-expressing the rcCymR-TAD.
(181) In some preferred embodiments, the cells of the cell line are present in a suspension. This has major advantages in the development of scalable and regulatory-accepted manufacturing processes, due to compatibility with stirred-tank and wave bioreactors, and through facilitating absence of animal components required for growth (e.g. FBS).
(182) The nucleic acid molecules, plasmids and vectors of the invention may be made by any suitable technique. Recombinant methods for the production of the nucleic acid molecules and packaging cells of the invention are well known in the art (e.g. “Molecular Cloning: A Laboratory Manual” (Fourth Edition), Green, M R and Sambrook, J., (updated 2014)).
(183) The expression of the rep and cap genes from the nucleic acid molecules of the invention may be assayed in any suitable assay, e.g. by assaying for the number of genome copies per ml by qPCR (as described the Examples herein).
(184) In a further embodiment, there is provided a process for producing an AAV cell line, the process comprising the steps of integrating into the genome of the cells of the cell line: (A) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, (ii) a cap gene, and (iii) a rep gene, wherein the rep gene preferably encodes Rep78 and Rep68, in the above 5′-3′ order, wherein the cap gene and the rep gene are both operably associated with the promoter, thereby producing a cell that inducibly expresses viral cap and rep genes; and optionally one of more of the following: (B) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter (preferably a minimal promoter) comprising one or more cumate operator sites, operably associated with (ii) a gene encoding a rc-CymR-TAD polypeptide; (C) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter (preferably a constitutive promoter), operably associated with (ii) a gene encoding CymR; (E) a nucleic acid molecule, vector or plasmid comprising: (i) one or more promoters, which may or may not comprise one or more cumate operator sites, operably associated with (ii) one or more adenoviral genes which are competent to support AAV production, preferably E4 and/or VAI-RNA.
(185) The nucleic acid molecules may, for example, be plasmids or vectors.
(186) In some preferred embodiments, the cells comprise nucleic acid molecule, vector or plasmid (A) and optionally nucleic acid, vector or plasmid (E). In these embodiments, the expression of the cap and rep genes and optionally adenoviral genes may be induced in the cell by the presence, in the cell (e.g. by transient transfection), of a CymR-TAD polypeptide. This binds to the cumate operator site(s), thus promoting transcription of the cap and rep genes.
(187) In other preferred embodiments, the cells comprise nucleic acid molecules, vectors or plasmids (A), (B), (C) and optionally nucleic acid, vector or plasmid (E). The expression of the cap and rep genes may be induced in the cell by the presence, in the cell, of cumate. This binds to the rcCymR-TAD, thus promoting transcription of the cap and rep genes. Cumate also binds to the CymR, thus preventing CymR from repressing transcription of the cap and rep genes.
(188) Preferably, the AAVs are subsequently harvested and/or purified.
(189) Each of the above nucleic acids, vectors and plasmids may be introduced into the cell. As used herein, the term “introducing” one or more nucleic acids, plasmids or vectors into the cell includes transformation, and any form of electroporation, conjugation, infection, transduction or transfection, inter alia. Processes for such introduction are well known in the art (e.g. Proc. Natl. Acad. Sci. USA. 1995 Aug. 1; 92 (16):7297-301).
(190) The transgene is flanked by inverted terminal repeats (ITRs). In some preferred embodiments, the transgene encodes a CRISPR enzyme (e.g. Cas9, Cpf1) or a CRISPR sgRNA.
(191) The invention also provides a method of inducing transcription of cap and rep genes in a cell, the method comprising the steps: inducing the transcription of cap and rep genes in a cell which comprises in its genome: (A) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, (ii) a cap gene, and (iii) a rep gene, wherein the rep gene preferably encodes Rep78 and Rep68, in the above 5′-3′ order, wherein the cap gene and the rep gene are both operably associated with the promoter, wherein the inducing is by the presence, in the cell, of a CymR-TAD polypeptide.
(192) The transcription of the cap and rep genes is induced in the cell by the presence, in the cell, of a

CymR-TAD polypeptide. This binds to the cumate operator site(s), thus promoting transcription of the cap and rep genes. The CymR-TAD may be introduced into the cell as a polypeptide. Preferably, however, the CymR-TAD is expressed in the cell, e.g. on a plasmid or vector (which may itself be introduced into the cell at a desired time). The CymR-TAD may be transiently expressed in the cell.
(193) Preferably, the cell line is a packaging cell line or a producer cell line.
(194) In some embodiments, the method additionally comprises the step of introducing: (C) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter (preferably a constitutive promoter), operably associated with (ii) a nucleotide sequence encoding a CymR-TAD; and/or (D) a nucleic acid molecule, vector or plasmid comprising: (i) one or more promoters, which may or may not comprise one or more cumate operator sites, operably associated with (ii) one or more adenoviral genes which are competent to support AAV production, preferably E4 and/or VAI-RNA; and/or (E) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter, operably associated with (ii) a transgene, wherein the promoter and transgene are flanked by ITRs, into the cell.
(195) The presence of CymR-TAD in the cell induces expression of the cap and rep genes.
(196) Nucleic acid molecules (C), (D) and (E) may be in the form of a plasmid or vector. In some embodiments, nucleic acid molecules (C) and (E) are on the same vector or plasmid. This therefore avoids the need for two separate transfection steps.
(197) The invention also provides a method of inducing transcription of cap and rep genes in a cell, the method comprising the steps: inducing the transcription of cap and rep genes in a cell which comprises in its genome: (A) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, (ii) a cap gene, and (iii) a rep gene, wherein the rep gene preferably encodes Rep78 and Rep68, in the above 5′-3′ order, wherein the cap gene and the rep gene are both operably associated with the promoter; (B) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, operably associated with (ii) a gene encoding a rc-CymR-TAD polypeptide; (C) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter, operably associated with (ii) a gene encoding CymR; and optionally (D) a nucleic acid molecule, vector or plasmid comprising: (i) one or more promoters, which may or may not comprise one or more cumate operator sites, operably associated with (ii) one or more adenoviral genes which are competent to support AAV production, preferably E4 and/or VAI-RNA; and/or (E) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter, operably associated with (ii) a transgene, wherein the promoter and transgene are flanked by ITRs, wherein (A), (B), (C), (D) and (E), when present, are preferably all integrated into the genome of the cell, wherein the inducing is by the presence, in the cell, of cumate.
(198) The expression of the cap and rep genes may be induced in the cell by the presence, in the cell, of cumate. In absence of cumate, the CymR binds to the promoter region of the rcCymR and represses transcription, and also binds to the minimal promoters driving the rep-cap (and/or Adenoviral) genes sterically blocking binding of residual rcCymR-TAD. In the presence of cumate, the repression of rcCymR-TAD transcription is relieved, as is the steric blocking, and the rcCymR-TAD is allowed to bind to the promoter of the cap-rep gene (and/or Adenoviral genes), and activates transcription—resulting in rAAV production. Cumate may be introduced into the cell by introducing cumate into the media surrounding the cell, at the desired time.
(199) Preferably, the cell is a producer cell line.
(200) The disclosure of each reference set forth herein is specifically incorporated herein by reference in its entirety.

EXAMPLES

(201) The present invention is further illustrated by the following Examples, in which parts and percentages are by weight and degrees are Celsius, unless otherwise stated. It should be understood that these Examples, while indicating preferred embodiments of the invention, are given by way of illustration only. From the above discussion and these Examples, one skilled in the art can ascertain the essential characteristics of this invention, and without departing from the spirit and scope thereof, can make various changes and modifications of the invention to adapt it to various usages and conditions. Thus, various modifications of the invention in addition to those shown and described herein will be apparent to those skilled in the art from the foregoing description. Such modifications are also intended to fall within the scope of the appended claims.

Example 1: CMV Promoter Comprising CuO Site

(202) Plasmid Design

(203) A CMV promoter containing a CuO site was designed. Two variants were created: the first had one CuO site; the second had a second CuO site downstream of the +1 site to increase steric blocking effects on transcription. The variants are shown in FIGS. **3**A and **3**B.

(204) The CMV-CuO promoter was cloned into Q414 (pSF-CMV-EGFP), resulting in Q5422 (pSF-CMV_CuO-EGFP), with CuO in the position shown in FIG. **3**A. A second CuO site was cloned into Q5422 resulting in Q5424 (pSF-CMV_CuOx2-EGFP), with the promoter and 5′UTR as shown in FIG. **3**B.

(205) The gene encoding the Cumate repressor protein (CymR—accession number: AB042509.1) was synthesized and cloned into OG10 (pSF-CMV-kan), resulting in Q4721 (pSF-CMV-CymR).

(206) Each of the above EGFP plasmids were analyzed by transfection in a 96-well plate format, with and without Q4721, and with and without cumate (30 μg/mL in 95% EtOH).

(207) Where Q4721 was omitted, OG10 was used as stuffer DNA. Where cumate was omitted, 95% EtOH was added.

(208) Transfection Setup

(209) Six DNA mixes were prepared as per Table 2. Transient transfections were carried out using a standard protocol for PEI transfection of Adherent cell lines, in HEK293 cells. DNA:PEI ratio 1:2.5 was used, with 0.14 μg total DNA transfected per well of 96 well plate. Plasmids were added in equimolar ratios, with each mix tested in triplicate.

(210) For the cumate solution, a stock solution of 5 mg/mL was prepared by dissolving cumate (Sigma, 268402) in 95% EtOH (Honeywell, 24103). 3 μL of either cumate stock solution or 95% EtOH was added to 197 μL DMEM+10% FBS. Prior to addition of DNA:PEI complexes to cells, 5 μL of either cumate-DMEM or EtOH-DMEM was added to each well (final cumate concentration of 30 μg/mL).

(211) 48 hours post-transfection, cells were prepared for flow cytometry analysis using the Attune NxT Flow Cytometer.

(212) TABLE-US-00001 TABLE 2 DNA mixes for transfections DNA mix EGFP plasmid Other 1 - CMV-CuO + CymR Q5422 Q4721 2 - CMV-CuO − CymR Q5422 OG10 3 - CMV-CuOx2 + CymR Q5424 Q4721 4 - CMV-CuOx2 − CymR Q5424 OG10 5 - CMV + CymR Q414 Q4721 6 - CMV − CymR Q414 OG10

Results

(213) FIG. **4** shows the Mean fluorescence Intensity of cells after 48 hours. Each biological triplicate was analyzed as a singlet, with 10,000 events counted from the live cell population. An EGFP negative sample was used to gate around non-fluorescent cells. The highest expression levels were observed for the constitutively-active CMV in Q414. The addition of cumate in these samples had a small impact on expression.

(214) For Q5422, with one CuO in the CMV promoter, repression was observed where CymR was present without cumate. Compared to Q5422+OG10+EtOH, this was calculated to be 5.6-fold repression (17.7% of the constitutive equivalent). The induction when CymR was present with cumate was equivalent to the constitutive control.

(215) As mirrored in the visual inspection of cells, Q5424 showed better repression than Q5422 (13.5% expression compared to the constitutive equivalent, 7.3-fold repression). The induction was lower, with 81% of the constitutive equivalent rather than the comparable expression seen with Q5422.

(216) Internal target specifications for the inducible system were set whereby repression should be <5% of the constitutive, and induction >75% of the constitutive. Neither Q5422 nor Q5424 appeared to yield sufficient repression when CymR was present without cumate to meet these criteria. They do both, however, show sufficient induction when cumate is added. Without CymR present, both CuO containing promoters showed constitutive activity. This meant that the system relied on constitutive and sufficient expression of CymR such that the inducible promoter remained off until cumate was added.

Example 2: CymR-Transactivators

(217) To resolve the issue of background expression of toxic proteins, a minimal promoter was designed which was basally inactive unless activated in some way. This minimal promoter can also be used with DNA-binding proteins fused to transcriptional activator domains (TADs, transactivators).

(218) Plasmid Design
(219) CymR-TAD fusion proteins were designed starting from Q4721 (pSF-CMV-CymR), the stop codon for CymR was removed and an oligo duplex of the TAD was cloned in-frame, with a new stop codon. To test additional TADs, several constructs were designed wherein TAD multimers were used. The resulting plasmids are listed in the table below. 'GS' in this instance refers to a glycine-serine linker motif.
(220) TABLE-US-00002 TABLE CymR-TAD plasmids generated Plasmid Identification Plasmid Name Q5622 CymR_VP16 Q5624 CymR_p53 Q5813 CymR_p53_VP16 Q5814 CymR_p53_p53 Q5866 CymR_VP16_GS
(221) The CuO-based minimal promoter used was with six CuO sites placed with small spacers upstream of the TATA box of the minimal CMV. This promoter was cloned upstream of an EGFP, resulting in Q5868 (pSF-6×CuO_minCMV-EGFP).
(222) This experiment was setup with positive promoter control conferred by Q414 (pSF-CMV-EGFP) and negative promoter control provided by Q437 (pSF-minCMV-EGFP). CymR-only plasmid Q4721 was not included, but a stuffer DNA control was included. This is because the key parameters are the promoter activity in the absence of any inducer (inducible EGFP+stuffer DNA) and the promoter activity in the presence of an inducer (inducible EGFP+CymR-TAD). Furthermore, the positive promoter control Q414 was not tested in combination with each CymR-TAD. It was anticipated that it would perform similarly with each.
(223) Transfection Setup
(224) Thirteen DNA mixes were prepared as per the table below. Transient transfections were carried out using a standard protocol for PEI transfection of Adherent cell lines, in HEK293 cells. DNA:PEI ratio 1:2.5 was used, with 0.14 μg total DNA transfected per well of 96 well plate. Plasmids were added in equimolar ratios, with each mix tested in triplicate. 48 hours post-transfection cells were prepared for flow cytometry analysis using the Attune NxT Flow Cytometer.
(225) TABLE-US-00003 TABLE DNA mixes for transfections DNA mix EGFP plasmid Other 1 - 6xCuO_minCMV-EGFP + Q5868 Q5622 CymR-VP16 2 - 6xCuO_minCMV-EGFP + Q5868 Q5624 CymR-p53 3 - 6xCuO_minCMV-EGFP + Q5868 Q5813 CymR-p53-VP16 4 - 6xCuO_minCMV-EGFP + Q5868 Q5814 CymR-p53x2 5 - 6xCuO_minCMV-EGFP + Q5868 Q5866 CymR-VP16_GS 6 - 6xCuO_minCMV-EGFP + OG10 Q5868 OG10 7 - minCMV-EGFP + CymR-VP16 Q437 Q5622 8 - minCMV-EGFP + CymR-p53 Q437 Q5624 9 - minCMV-EGFP + CymR-p53-VP16 Q437 Q5813 10 - minCMV-EGFP + CymR-p53x2 Q437 Q5814 11 - minCMV-EGFP + CymR-VP16_GS Q437 Q5866 12 - minCMV-EGFP + OG10 Q437 OG10 13 - CMV-EGFP + OG10 Q414 OG10
Results
(226) FIG. **5** shows the mean fluorescence intensity as measured by flow cytometry. The first observation was that when there was a CymR-TAD present with the inducible EGFP plasmid, fluorescence was higher than with the positive promoter control (CMV-EGFP). This was unexpected; however, this was observed with CymR-TADs in subsequent experiments.
(227) As expected, the negative promoter control (minCMV) showed very low basal activity. In this instance between 0.95% and 2.25% activation compared to the positive control (CMV). Comparatively low basal expression was seen with 6×CuO_minCMV-EGFP in the absence of induction: at 1.19% of the constitutive. This confirms that the promoter is transcriptionally inactive without inducers present and behaves as the negative control minCMV.
(228) As all the CymR-TAs induced beyond the level of the constitutive control, it was concluded that all surpassed the 75% activation threshold. Additionally, conclusions were drawn about the relative transcriptional activation strength of each TAD or TAD-multimer. This data showed that the presence of VP16 conferred the highest activation, and the addition of a p53 TAD to the VP16 TAD gave no additional transcriptional activation activity. Similarly, the presence of a glycine-serine linker did not hinder the transcriptional activity of VP16. This meant that VP16 domains can be linked with linkers with minimal impact on function. On the other hand, p53 TAD did not appear to activate transcription as strongly as VP16. Two p53 TADs concurrently did not improve the activity.
(229) In an identically set up experiment, a further set of TADs were compared, shown in FIG. **6**. Experiment setup was identical as described in this Example, with the exception that the negative promoter minCMV control was not run in parallel—it was assumed that expression from this promoter

would be negligibly low (as in FIG. **5**). FIG. **6** shows again how the CymR-TADs allowed EGFP expression to exceed that driven by the constitutive CMV promoter and allowed comparison of two previously-uncharacterized TADs: VP16_p53 and VP32 (VP16-VP16). VP32 appeared to be the strongest TAD in this data, with p53 alone performing most weakly.
(230) As previously observed, this data showed <5% expression from the uninduced inducible promoter (2.75% from 6×CuO_minCMV-EGFP+OG10 compared to the constitutive CMV-EGFP+OG10). Additionally, there was over 100% activation upon induction when compared to the constitutive.
Example 3: Further CymR-Transactivators
(231) After initial testing of CymR-TADs using the simple EGFP expression experiment, the system was tested in the context of rAAV production. The CMV promoter in Q5220 (pSF-CMV-Cap5-EMCV-Rep) was replaced with the inducible 6×CuO_minCMV promoter resulting in Q5875 (pSF-6×CuO_minCMV-Cap5-EMCV-Rep).
(232) Virus production was carried out using either the inducible packaging plasmid Q5875 or constitutive Q5220, standard helper plasmid Q1364 and standard EGFP transgene Q4346. In each instance, the CymR-TAD was delivered as for the EGFP experiment in Example 3: in the first rAAV experiment, only CymR-VP16 was tested, but in a subsequent experiment three other CymR-TADs were evaluated in their ability to induce viral production using the inducible promoter.
(233) Transfection Setup
(234) Five mixes of DNA were prepared for the first experiment, and seven for the second, as per the table below. Transient transfections were carried out with plasmids added in equimolar ratios. For each distinct DNA mix, three wells of a 6-well plate of adherent HEK293s were transfected. A total DNA mass of 2.5 μg was transfected per well. 72 hours post-transfection, the contents of each well was harvested using chemical lysis and crude AAV lysate obtained.
(235) TABLE-US-00004 TABLE DNA mixes for transfections Transgene Packaging Helper Expt DNA mix Plasmid Plasmid Plasmid Other 1 1 - Inducible RepCap + Q4346 Q5875 Q1364 Q5622 CymR-VP16 2 - Inducible RepCap Q4346 Q5875 Q1364 OG10 3 - Constitutive RepCap + Q4346 Q5220 Q1364 Q5622 CymR-VP16 4 - Constitutive RepCap Q4346 Q5220 Q1364 OG10 5 - RepCap5 production Q4346 Q4658 Q1364 N/A control 2 1 - Inducible RepCap + Q4346 Q5875 Q1364 Q5974 CymR-VP16-p53 2 - Inducible RepCap + Q4346 Q5875 Q1364 Q5973 CymR-VP32 3 - Inducible RepCap + Q4346 Q5875 Q1364 Q5624 CymR-p53 4 - Inducible RepCap + Q4346 Q5875 Q1364 Q4721 CymR 5 - Inducible RepCap Q4346 Q5875 Q1364 OG10 6 - Constitutive RepCap Q4346 Q5220 Q1364 OG10 7 - RepCap5 production Q4346 Q4658 Q1364 N/A control
Results
(236) Crude AAV samples were analyzed for titre values by qPCR, to ascertain if virus production could be controlled using the 6×CuO_minCMV promoter driving the RepCap cassette and with addition or absence of a CymR-TAD. The primers for qPCR targeted the UTR of the transgene plasmid, with plasmid Q4346 used as the standard in the assay. Vector genomes were calculated from qPCR data using standard approaches.
(237) FIG. **7** shows the viral genomes per mL from the first experiment. With no added activator, there was 2.35% activation compare to the constitutive equivalent packaging plasmid. This fell below the 5% threshold set, mirroring the data from the EGFP experiments. When activated (CymR-VP16 present), there was 78.5% viral production compared to the constitutive (if taking the constitutive+OG10). If comparing to the constitutive equivalent (+CymR-VP16) the viral production was over 100%. This would be consistent with the observations in the EGFP experiment also. In both scenarios however, observed promoter activation was over the target 75% threshold.
(238) FIG. **8** shows the viral genomes per cell, or cell specific productivity. For a viable packaging cell line, the ideal cell specific productivity (for an induced cell line) is >25,000 VG/cell, with <10,000 being the bottom cut-off. None of the samples in this experiment met the ideal 25,000 viral genomes per cell; however, the induced system was over the 10,000 VG/cell.
Example 4: rcCymR-Transactivators
(239) In order to avoid using a DNA-based method of induction in a fully stable producer cell line, we investigated the use of a CymR mutant, rc-CymR, which can bind DNA only in the presence of cumate. By employing the same TAD-domain based transcriptional activation method, it may be possible to use a

cumate-activated rc-CymR-TAD to regulate rAAV production.
(240) The system would be used as follows: all AAV DNA genetic elements (inducible Rep and Cap genes, Ad5 helper genes, ITR-flanked transgene) would be stably integrated into the cell line, along with a constitutively expressed rc-CymR-TA. The rc-CymR-TAD protein would be unable to bind DNA unless cumate was present, preventing the TAD-mediated transcriptional activation. To induce rAAV production, cumate can be added, allowing the rc-CymR-TAD to bind the inducible promoters and initiate transcription of the AAV packaging genes and thus allowing viral production.
(241) Plasmid Design
(242) Three point mutations were introduced at amino acid positions A122V, E139G and M141I within the CymR sequence. This resulted in plasmid Q5812 (pSF-CMV-rc-CymR). From this, p53 and VP16 TADs were fused, resulting in Q5873 (pSF-CMV-rc-CymR-p53) and Q5874 (pSF-CMV-rc-CymR-VP16).
(243) These plasmids were first tested with the inducible EGFP plasmid Q5868 (pSF-6×CuO_minCMV-EGFP) to assess whether or not the functionality with relation to cumate had changed.
(244) Transfection Setup
(245) Nine DNA mixes were prepared as in the table below. Transient transfections were carried out using a standard protocol for PEI transfection of Adherent cell lines, in HEK293 cells. DNA:PEI ratio 1:2.5 was used, with 0.14 μg total DNA transfected per well of 96 well plate. Plasmids were added in equimolar ratios, with each mix tested in triplicate.
(246) For the cumate solution, a stock solution of 5 mg/mL was prepared by dissolving cumate (Sigma, 268402) in 95% EtOH (Honeywell, 24103). 3 μL of either cumate stock solution or 95% EtOH was added to 197 μL DMEM+10% FBS. Prior to addition of DNA:PEI complexes to cells, 5 μL of either cumate-DMEM or EtOH-DMEM was added to each well (final cumate concentration of 30 μg/mL). 72 hours post-transfection cells were prepared for flow cytometry analysis using the Attune NxT Flow Cytometer.
(247) TABLE-US-00005 TABLE DNA mixes for transfections DNA mix EGFP plasmid Other 1 - 6xCuO_minCMV-EGFP + rc-CymR-VP16 Q5868 Q5874 2 - 6xCuO_minCMV-EGFP + rc-CymR-p53 Q5868 Q5873 3 - 6xCuO_minCMV-EGFP + rc-CymR Q5868 Q5812 4 - 6xCuO_minCMV-EGFP Q5868 OG10 5 - minCMV-EGFP + rc-CymR-VP16 Q437 Q5874 6 - minCMV-EGFP + rc-CymR-p53 Q437 Q5873 7 - minCMV-EGFP + rc-CymR Q437 Q5812 8 - minCMV-EGFP Q437 OG10 9 - CMV-EGFP Q414 OG10
Results
(248) FIG. **9** shows the mean fluorescence intensity after 72 hours. It is immediately clear that as observed previously, there was minimal background EGFP expression resulting from either the minimal CMV or the 6×CuO_minCMV promoter without induction. There was also no significant difference between the constitutive samples with and without cumate.
(249) What was hypothesized was that with the inducible EGFPF+rc-CymR-TAD, EGFP expression would be observed with cumate but not without. However, although there was significantly higher EGFP expression with cumate than without, there was still significant observable EGFP expression without cumate. For rc-CymR-VP16, this amounted to approximately 19-fold more expression than the equivalent negative control. For rc-CymR-p53 this value is even higher, at over 100-fold more.
(250) Upon induction with cumate, the EGFP expression was not as high as anticipated. For rc-CymR-VP16, induced EGFP was 7.3% of the constitutive, whilst rc-CymR-p53 performed better, at 37%. Neither, however, were close to the ideal 75% induction compared to the constitutive.
(251) Overall, although rc-CymR was showing altered DNA binding properties with relation to cumate compared to CymR, there was high residual binding without cumate present. In a stable cell line context, this would lead to high basal expression of cytotoxic Rep proteins, and poor control upon induction of the system with cumate.
(252) Corresponding experiments were performed in the context of rAAV production and similar results were obtained.
(253) Significant rc-CymR-VP16 binding was observed in the absence of cumate. In this instance, the inducible RepCap showed 2.5% viral production compared to the constitutive, when no rc-CymR-VP16 was present. However, when rc-CymR-VP16 was present, both with and without cumate there was no significant difference to the constitutive expression. Furthermore, there was very high rAAV production

when no cumate was present. This indicated that the induction system would not work in a stable cell line in this form.

Example 5: Dual Control System

(254) A new strategy was designed to reduce background rc-CymR DNA binding in the absence of cumate. This system involves what is known as ‘dual cumate control’, whereby there are two concurrent systems operating to prevent transcription of the target gene, which are both relieved by addition of cumate.

(255) FIGS. **1**A and **1**B show schematics for how this system operates with and without cumate. For simplicity, the target regulated gene is EGFP; in a producer cell line, it would be the regulated AAV packaging elements. Three core DNA components of the system are required: a constitutively expressed CymR protein; a CymR-repressible rc-CymR-TAD protein; the target regulated gene driven by the 6×CuO_minCMV promoter (requiring CymR-TAD or rc-CymR-TAD binding for activation).

(256) In the absence of cumate, only the CymR is expressed and this has two functions: First, the CymR binds to the CuO in the rc-CymR-TAD promoter, repressing transcription of rc-CymR-TAD—thereby preventing the background binding that was observed when rc-CymR was used alone. Second, CymR will bind to the 6×CuO_minCMV promoter and sterically block any rc-CymR-TAD that has been transcribed through leakiness of the CymR-repression. The constitutive expression of CymR is likely to greatly out-compete any rc-CymR-TAD for binding at these 6×CuO sites—thereby the target regulated gene is kept off.

(257) When cumate is added, the CymR can no longer bind to DNA. This will remove the steric block at the 6×CuO sites and relieve repression of the rc-CymR-TAD promoter. Furthermore, the cumate should fully activate DNA binding abilities of rc-CymR-TAD—allowing the transcriptional activation of the target regulated gene through the single addition of cumate.

(258) In a pilot experiment, EGFP expression from an inducible plasmid (Q5868, pSF-6×CuO_minCMV-EGFP) was tested using this dual cumate control system.

(259) Plasmid Design

(260) Most of the plasmids needed to test this system using an inducible EGFP were already available, with the exception of the repressible rc-CymR-TAD. This was cloned by placing a CMV_CuO promoter upstream of a rc-CymR-VP16 coding sequence, resulting in Q6440 (pSF-CMV_CuO-rc-CymR-VP16).

(261) Transfection Setup

(262) Six DNA mixes were prepared as per the table below. Transient transfections were carried out using a standard protocol for PEI transfection of Adherent cell lines, in HEK293 cells. DNA:PEI ratio 1:2.5 was used, with 0.14 μg total DNA transfected per well of 96 well plate. Plasmids were added in equimolar ratios, with each mix tested in triplicate.

(263) For the cumate solution, a stock solution of 5 mg/mL was prepared by dissolving cumate (Sigma, 268402) in 95% EtOH (Honeywell, 24103). 3 μL of either cumate stock solution or 95% EtOH was added to 197 μL DMEM+10% FBS. Prior to addition of DNA:PEI complexes to cells, 5 μL of either cumate-DMEM or EtOH-DMEM was added to each well (final cumate concentration of 30 μg/mL).

(264) TABLE-US-00006 TABLE DNA mixes for transfections EGFP DNA mix plasmid Other 1 - 6xCuO_minCMV-EGFP + CMV_CuO- Q5868 Q6440 Q4721 rc-CymR-VP16 + CymR 2 - 6xCuO_minCMV-EGFP + CMV_CuO- Q5868 Q6440 T176 rc-CymR-VP16 3 - 6xCuO_minCMV-EGFP + CMV-rc- Q5868 Q5874 Q4721 CymR-VP16 + CymR 4 - 6xCuO_minCMV-EGFP + CMV- Q5868 Q5874 T176 rc-CymR-VP16 5 - minCMV-EGFP +CMV-rc-CymR- Q437 Q5874 Q4721 VP16 + CymR 6 - CMV-EGFP + CMV-rc-CymR- Q414 Q5874 Q4721 VP16 + CymR

Results

(265) FIGS. **10**A and **10**B show the flow cytometry data for samples after 72 hours. From the mean fluorescence intensity, it was apparent that the dual control system has restored stringency to the system of using cumate to activated rc-CymR-TAD DNA binding as the activation of the target regulated gene (EGFP).

(266) Where the system was fully present (6×CuO_minCMV-EGFP+CMV_CuO-rc-CymR-VP16+CMV-CymR), there were significant differences between the samples with cumate and without. Without cumate, with the system intended to be “off”, there was 7.4% fluorescence compared to the constitutive, and just 1.6-fold more than the negative minCMV control. Conversely, upon addition of cumate, there

was over 100% fluorescence compared to the constitutive, with induction giving a 17-fold increase.
(267) As previously observed above, when no CMV-CymR was present (6×CuO_minCMV-EGFP+CMV_CuO-rc-CymR-VP16) there was little difference between samples with and without cumate. This correlated with previous observations that there was high rc-CymR DNA binding even in the absence of cumate. In this scenario, the CMV CuO was basally active, unless repressed by CymR protein.
(268) The dual effect of the control was observable when the rc-CymR-VP16 was not regulated, but CMV-CymR was present (6×CuO_minCMV-EGFP+CMV-rc-CymR-VP16+CMV-CymR). The rc-CymR-VP16 was not being repressed by CymR, but it was facing competition for binding at the 6×CuO sites. Without cumate, when CymR was able to bind CuO sites, there was significantly lower EGFP intensity than the induced when CymR was prevented from binding due to presence of cumate. The repression was 16.9% of the constitutive equivalent. Upon induction, EGFP fluorescence intensity increased over 5-fold.
(269) There was no significant difference between samples with and without cumate for the negative promoter control and positive promoter control (minCMV-EGFP and CMV-EGFP respectively).
Example 6: Reverse Cumate Dual Control System Using CymR and Rc_CymR-VP16 Present on a Transgene Cassette Plasmid
(270) CymR and rc_CymR-VP16 were cloned from Q4721 (pSF-CMV-CymR) and Q6440 (pSF-CMV-CuO-rc_CymR-VP19 respectively into the vector backbone of Q6794 (pSF-AAV-ITR-CMV-EGFP-ITR), generating Q9003 (pSF-AAV-CMV-CuO-rc_CymR-VP16-CMV-EGFP-CMV-CymRSalIMluI).
(271) 293 baseline suspension cells were transfected by PEIPro (1 μg DNA:2p1 PEIPro) in 24 deep well format, 3 μg DNA per well. Enough transfection mix was prepared for four well per test plasmid condition, giving 2 vehicle control wells and 2 cumate treated wells. Final concentration of cumate added to each well was 30 μg/ml, with the equivalent volume of 95% ethanol (EtOH) applied to vehicle control wells. Treatment applied to each well by addition of 100 μl BalanCD media after application of transfection mix. Positive control transfections for standard AA5 production (Q6794+Q5220+Q1364) were also performed alongside test conditions as well as the negative controls: no genome (OG590+Q5220+Q1364) and no RepCap (Q6794+OG590+Q1364). OG590 (pUC19) used as stuffer in control transfections using the equivalent volume of DNA as the positive control in each instance.
(272) Q7981 is pSF-AAV-6×CuO-Cap5-EMCV-Rep-E4VAI-UB-IoxP-Puro-IoxP.
(273) Q7951 is pSF-AAV-6×CuO-Cap5-EMCV-Rep-6×CuO-E4VAI-UB-Puro.
(274) Q6794 pSF-ITR-CMV-EGFP-ITR.
(275) TABLE-US-00007 24DWP Well # Experiment Description Plasmid codes 1-2 All-in-one transactivator + inducible Q9003 + Q7981 Cap + Vehicle control 3-4 All-in-one transactivator + inducible Q9003 + Q7981 Cap + Cumate 5-6 All-in-one transactivator + inducible Q9003 + Q7951 Cap/inducible VAIE4 + Vehicle control 7-8 All-in-one transactivator + inducible Q9003 + Q7951 Cap/inducible VAIE4 + Cumate 9-10 Triple Transfection Q6794 + Q1364 + Q5220 11-12 No RepCap control Q6794 + Q1364 + OG590 13-14 No genome control OG590 + Q5220 + Q1364 15-16 EGFP control Q414 17 Cumate control — 18 Vehicle control (95% EtoH) —
(276) Cells were harvested 3 days post transfection by standard chemical lysate. Samples were process for qPCR using the standard DNaseI and proteinase K treatment. qPCR performed using the UTR primer/probe mix with linear Q6794 used as a serially-diluted standard for quantification of sample viral genomes (VG)/ml.
(277) The results are shown in FIGS. **11-12**. In the absence of cumate, basal virus production was <2.5% of cumate-induced production. Cumate-induced production yielded 27-31% of the VG/ml compared to standard triple transfection. Data is shown as the average VG/ml from the two biological replicates from the single transfection; errors bars are for the combined standard error.
Example 7: Validation of CymR-Transactivators in rAAV Production in Suspension HEK293 Cells
(278) Validation of the proposed induction system for a stable packaging cell line involved a screen of all CymR-TAD plasmids with various “all-in-one” plasmids.
(279) Experiment Design and Setup
(280) 24 hours prior to transfection, two 24 deep-well plates were seeded with 1.0E+06 cells/mL of HEK 293-OX (parental cell line) with a total of 3 mL per well, in BalanCD media. Cells were seeded at passage number 14. Fourteen DNA mixes were set up according to the Table below.
(281) TABLE-US-00008 TABLE DNA mixes for transfections Transgene Packaging Other DNA mix

Plasmid Plasmid Plasmid 1 Inducible RepCap CymR-VP16 Q4346 Q6297 Q5622 2 Inducible RepCap CymR-p53 Q4346 Q6297 Q5624 3 Inducible RepCap Cym R-VP32 Q4346 Q6297 Q5973 4 Inducible RepCap CymR- Q4346 Q6297 Q5814 p53x2 5 Inducible RepCap CymR- Q4346 Q6297 Q5974 VP16-p53 6 Inducible RepCap CymR-p53- Q4346 Q6297 Q5813 VP16 7 Inducible RepCap CymR- Q4346 Q6297 Q5866 VP16-GS 8 Inducible RepCap N/A Q4346 Q6297 T176 9 Constitutive RepCap N/A Q4346 Q6295 N/A 10 RepCap 5 production N/A Q4346 Q4658 Q1364 control 1 11 RepCap 5 production N/A Q4346 Q5220 Q1364 control 2 12 No RepCap control N/A Q4346 T176 Q1364 13 No Genome control N/A T176 Q5220 Q1364 14 EGFP Transfection N/A N/A N/A Q414 control

Results

(282) Crude AAV samples were analyzed for titre values by qPCR, to ascertain how the inducible plasmid performs in a suspension HEK293 system, and to compare each CymR-TAD against one another. The primers for qPCR targeted the UTR of the transgene plasmid, with plasmid Q4346 used as the standard.

(283) FIG. **13** shows the viral genomes per mL. Immediately apparent is that with no transactivator there is very low viral production (1.75E+08 VG/m L). This corresponds to 0.8% of the constitutive equivalent; this shows very stringent repression. This result shows that the inducible promoter is sufficiently inactive, in a suspension HEK293 cells.

(284) The constitutive “re-configured” control (CMV-Cap5-EMCV-Rep control) is the highest performer. This is not surprising as it has superior helper functions supplied by Q1364 (containing E2A, E4 and VAI regions). Similarly, Q4658 (pSF-RepCap5) also was tested in combination with Q1364. These two served as production controls to ensure that production itself is within the expected range. Therefore, these two values cannot be compared to the rest of the data—but indicate that the experimental setup was a success. Similarly, the no-genome and no-repcap controls are not depicted on the graphs as they fell outside the range of the standard curve and can be considered negligible.

(285) Comparing each CymR-TA, we see that the top three are CymR-VP16, CymR-p53 and CymR-VP16_GS. These give 95%, 82% and 70% viral titre compared to the constitutive.

(286) Overall, these data demonstrate that the proposed induction system for use in a stable AAV packaging cell line meets both criteria for repression and activation in suspension HEK293 cells. Less than 1% viral production is observed in the “off” state, and up to 95% activation is seen upon addition of the DNA activator CymR-TA.

(287) TABLE-US-00009 SEQUENCES Rep nucleotide sequence (AAV serotype 2) SEQ ID NO: 1 atgccggggttttacgagattgtgattaaggtccccagcgaccttgacgagcatctgcccggca
tttctgacagctttgtgaactgggtggccgagaaggaatgggagttgccgccagattctgacat
ggatctgaatctgattgagcaggcacccctgaccgtggccgagaagctgcagcgcgactttctg
acggaatggcgccgtgtgagtaaggccccggaggcccttttctttgtgcaatttgagaagggag
agagctacttccacatgcacgtgctcgtggaaaccaccggggtgaaatccatggttttgggacg
tttcctgagtcagattcgcgaaaaactgattcagagaatttaccgcgggatcgagccgactttg
ccaaactggttcgcggtcacaaagaccagaaatggcgccggaggcgggaacaaggtggtggatg
agtgctacatccccaattacttgctccccaaaacccagcctgagctccagtgggcgtggactaa
tatggaacagtatttaagcgcctgtttgaatctcacggagcgtaaacggttggtggcgcagcat
ctgacgcacgtgtcgcagacgcaggagcagaacaaagagaatcagaatcccaattctgatgcgc
cggtgatcagatcaaaaacttcagccaggtacatggagctggtcgggtggctcgtggacaaggg
gattacctcggagaagcagtggatccaggaggaccaggcctcatacatctccttcaatgcggcc
tccaactcgcggtcccaaatcaaggctgccttggacaatgcgggaaagattatgagcctgacta
aaaccgcccccgactacctggtgggccagcagcccgtggaggacatttccagcaatcggattta
taaaattttggaactaaacgggtacgatccccaatatgcggcttccgtctttctgggatgggcc
acgaaaaagttcggcaagaggaacaccatctggctgtttgggcctgcaactaccgggaagacca
acatcgcggaggccatagcccacactgtgcccttctacgggtgcgtaaactggaccaatgagaa
ctttcccttcaacgactgtgtcgacaagatggtgatctggtgggaggaggggaagatgaccgcc
aaggtcgtggagtcggccaaagccattctcggaggaagcaaggtgcgcgtggaccagaaatgca
agtcctcggcccagatagacccgactcccgtgatcgtcacctccaacaccaacatgtgcgccgt
gattgacgggaactcaacgaccttcgaacaccagcagccgttgcaagaccggatgttcaaattt
gaactcacccgccgtctggatcatgactttgggaaggtcaccaagcaggaagtcaaagactttt
tccggtgggcaaaggatcacgtggttgaggtggagcatgaattctacgtcaaaaagggtggagc

caagaaaagacccgcccccagtgacgcagatataagtgagcccaaacgggtgcgcgagtcagtt
gcgcagccatcgacgtcagacgcggaagcttcgatcaactacgcagacaggtaccaaaacaaat
gttctcgtcacgtgggcatgaatctgatgctgtttccctgcagacaatgcgagagaatgaatca
gaattcaaatatctgcttcactcacggacagaaagactgtttagagtgctttcccgtgtcagaa
tctcaacccgtttctgtcgtcaaaaaggcgtatcagaaactgtgctacattcatcatatcatgg
gaaaggtgccagacgcttgcactgcctgcgatctggtcaatgtggatttggatgactgcatctt tgaacaaTAG Rep78 amino acid sequence (AAV serotype 2) SEQ ID NO: 2
MPGFYEIVIKVPSDLDGHLPGISDSFVNWVAEKEWELPPDSDMDLNLIEQAPLTVAEKLQRDFL
TEWRRVSKAPEALFFVQFEKGESYFHMHVLVETTGVKSMVLGRFLSQIREKLIQRIYRGIEPTL
PNWFAVTKTRNGAGGGNKVVDECYIPNYLLPKTQPELQWAWTNMEQYLSACLNLTERKRLVAQH
LTHVSQTQEQNKENQNPNSDAPVIRSKTSARYMELVGWLVDKGITSEKQWIQEDQASYISFNAA
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CGAATGCCCTCCGCCGCTTTGGGGCCACTGCTACCTTCTGCAGCTAGCCAACTACCTTGCCTAC
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GTGTCAGACGAAACACCGTCACCCTCGGTCGCATTCCCCTCGCCGGCGCCCCAGAAATCGGCAA CCGGTTCCAGCATGGCTACAACCTCCGCTCCTCAGGCGCCGCCGGCACTGCCCGTTCGCCGACC CAACCGTAGATGGGACACCACTGGAACCAGGGCCGGTAAGTCCAAGCAGCCGCCGCCGTTAGCC CAAGAGCAACAACAGCGCCAAGGCTACCGCTCATGGCGCGGGCACAAGAACGCCATAGTTGCTT GCTTGCAAGACTGTGGGGGCAACATCTCCTTCGCCCGCCGCTTTCTTCTCTACCATCACGGCGT GGCCTTCCCCCGTAACATCCTGCATTACTACCGTCATCTCTACAGCCCATACTGCACCGGCGGC AGCGGCAGCAACAGCAGCGGCCACACAGAAGCAAAGGCGACCGGATAGCAAGACTCTGACAAAG CCCAAGAAATCCACAGCGGCGGCAGCAGCAGGAGGAGGAGCGCTGCGTCTGGCGCCCAACGAAC CCGTATCGACCCGCGAGCTTAGAAACAGGATTTTTCCCACTCTGTATGCTATATTTCAACAGAG CAGGGGCCAAGAACAAGAGCTGAAAATAAAAAACAGGTCTCTGCGATCCCTCACCCGCAGCTGC CTGTATCACAAAAGCGAAGATCAGCTTCGGCGCACGCTGGAAGACGCGGAGGCTCTCTTCAGTA AATACTGCGCGCTGACTCTTAAGGACTAGTTTCGCGCCCTTTCTCAAATTTAAGCGCGAAAACT ACGTCATCTCCAGCGGCCACACCCGGCGCCAGCACCTGTTGTCAGCGCCATTATGAGCAAGGAA ATTCCCACGCCCTACATGTGGAGTTACCAGCCACAAATGGGACTTGCGGCTGGAGCTGCCCAAG ACTACTCAACCCGAATAAACTACATGAGCGCGGACCCCACATGATATCCCGGGTCAACGGAAT ACGCGCCCACCGAAACCGAATTCCCTTGGAACAGGCGGCTATTACCACCACACCTCGTAATAAC CTTAATCCCCGTAGTTGGCCCGCTGCCCTGGTGTACCAGGAAAGTCCCGCTCCCACCACTGTGG TACTTCCCAGAGACGCCCAGGCCGAAGTTCAGATGACTAACTCAGGGGCGCAGCTTGCGGGCGG CTTTCGTCACAGGGTGCGGTCGCCCGGGC CymR *Pseudomonas putida* SEQ ID NO: 17 atgagtccaaagagaagaacacaggcagagcgcgcaatggagacccagggcaagttgattgcag cggccctgggggttttacgggaaaaggttacgcgggattccggatcgcagatgtgcccggtgc tgcaggtgtctcgagaggagcgcagagccatcatttcccgacaaagcttgagcttctgcttgcc acttttgaatggctttacgaacagatcaccgaacgcagtcgggctcgattagcgaaattgaagc cagaggatgacgtcatccagcaaatgctggacgacgccgccgaatttttcctcgacgatgactt ctctatcagccttgatttgattgtggctgccgaccgggatccagcgttacgcgagggtattcag cgcacggtagagaggaatcggtttgtcgtcgaggatatgtggcttggtgttctggtgagccgtg gtctttcgcgtgatgatgcagaagatatcctttggttgatattcaattcggtgcgtgggcttgc tgttcgtagcctatggcagaaggacaaagaacgctttgagcgtgtcaggaactcgacactcgaa attgcgcgagagcggtacgcgaaattcaagcgctag CymR Protein *Pseudomonas putida* SEQ ID NO: 18 MSPKRRTQAERAMETQGKLIAAALGVLREKGYAGFRIADVPGAAGVSRGAQSHHFPTKLE LLLATFEWLYEQITERSRARLAKLKPEDDVIQQMLDDAAEFFLDDDFSISLDLIVAADRD PALREGIQRTVERNRFVVEDMWLGVLVSRGLSRDDAEDILWLIFNSVRGLAVRSLWQKDK ERFERVRNSTLEIARERYAKFKR rcCymR (In the following sequence, the mutations compared to CymR are highlighted.) SEQ ID NO: 19 atgagtccaaagagaagaacacaggcagagcgcgcaatggagacccagggcaagttgattgcag cggccctgggggttttacgggaaaaggttacgcgggattccggatcgcagatgtgcccggtgc tgcaggtgtctcgagaggagcgcagagccatcatttcccgacaaagcttgagcttctgcttgcc acttttgaatggctttacgaacagatcaccgaacgcagtcgggctcgattagcgaaattgaagc cagaggatgacgtcatccagcaaatgctggacgacgccgccgaatttttcctcgacgatgactt ctctatcagccttgatttgattgtggctgccgaccgggatcca**GTC**ttacgcgagggtattcag cgcacggtagagaggaatcggtttgtcgtc**GGC**gat**ATC**tggcttggtgttctggtgagccgtg gtctttcgcgtgatgatgcagaagatatcctttggttgatattcaattcggtgcgtgggcttgc tgttcgtagcctatggcagaaggacaaagaacgctttgagcgtgtcaggaactcgacactcgaa attgcgcgagagcggtacgcgaaattcaagcgctag rcCymR protein SEQ ID NO: 20 MSPKRRTQAERAMETQGKLIAAALGVLREKGYAGFRIADVPGAAGVSRGAQSHHFPTKLE LLLATFEWLYEQITERSRARLAKLKPEDDVIQQMLDDAAEFFLDDDFSISLDLIVAADRD PVLREGIQRTVERNRFVVGDIWLGVLVSRGLSRDDAEDILWLIFNSVRGLAVRSLWQKDK ERFERVRNSTLEIARERYAKFKR FIG. 3A sequence SEQ ID NO: 21 GGCGTGTACGGTGGGAGGTCTATATAAGCAGAGCTGaacaaacagacaatctggtctgtttgta GTTTAGTGAACCGAGATCTTTGTCGATCCT FIG. 3B sequence SEQ ID NO: 22 GGCGTGTACGGTGGGAGGTCTATATAAGCAGAGCTGaacaaacagacaatctggtctgtttgta GTTTAGTGAACCGAGATCTTTGTCGATCCTACCATCCACTCGACACACCCGCCAGCGGCCGCaa caaacagacaatctggtctgtttgtaAGCTICCGAGCTCTCGAATTCAAAGGAGGTACCCACCA TG

Claims

1. A nucleic acid molecule comprising: (i) a minimal promoter comprising 1, 2, 3, 4, 5 or 6 cumate operator (CuO) sites, (ii) a cap gene, and (iii) a rep gene, in the above 5′-3′ order, wherein the cap gene and the rep gene are both operably-associated with the promoter.
2. The nucleic acid molecule as claimed in claim 1, wherein the promoter is a minimal pol II promoter.
3. The nucleic acid molecule as claimed in claim 1, wherein the promoter comprises 4, 5 or 6 CuO sites.
4. The nucleic acid molecule as claimed in claim 1, wherein the CuO site(s) in the promoter are placed upstream of both the TATA box and the +1 site.
5. The nucleic acid molecule as claimed in claim 1, wherein the rep gene encodes Rep78 and Rep68, but not Rep 52 or Rep 40.
6. The nucleic acid molecule as claimed in claim 1, wherein the rep gene is operably-associated with an Internal Ribosome Entry Site (IRES) which is placed between the cap and rep genes.
7. The nucleic acid molecule as claimed in claim 1, wherein the rep gene encodes Rep78 and Rep68.
8. The nucleic acid molecule as claimed in claim 1, wherein the promoter comprises 6 CuO sites.
9. A vector or plasmid comprising the nucleic acid molecule of claim 1.
10. A kit comprising: (A) a nucleic acid molecule as claimed in claim 1; and optionally one or more of the following: (B) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter or a minimal promoter comprising one or more cumate operator sites, operably-associated with (ii) a gene encoding a rc-CymR-TAD polypeptide; (C) a nucleic acid molecule, vector or plasmid comprising (i) a promoter or a constitutive promoter, operably-associated with (ii) a gene encoding CymR; (D) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter or a constitutive promoter, operably-associated with (ii) a gene encoding a CymR-TAD polypeptide; (E) a nucleic acid molecule, vector or plasmid comprising: (i) one or more promoters, which may or may not comprise one or more cumate operator sites, operably-associated with (ii) one or more adenoviral genes which are competent to support AAV production; (F) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter, operably-associated with (ii) a transgene, wherein the promoter and transgene are flanked by ITRs.
11. A cell line, comprising: (A) a nucleic acid molecule as claimed in claim 1; and optionally one or more of the following: (B) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter or a minimal promoter comprising one or more cumate operator sites, operably-associated with (ii) a gene encoding a rc-CymR-TAD polypeptide; (C) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter or a constitutive promoter, operably-associated with (ii) a gene encoding CymR; (D) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter or a constitutive promoter, operably-associated with (ii) a gene encoding a CymR-TAD polypeptide; (E) a nucleic acid molecule, vector or plasmid comprising: (i) one or more promoters, which may or may not comprise one or more cumate operator sites, operably-associated with (ii) one or more adenoviral genes which are competent to support AAV production; (F) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter, operably-associated with (ii) a transgene, wherein the promoter and transgene are flanked by ITRs.
12. The cell line as claimed in claim 11, wherein the cell line is a packaging cell line or a producer cell line.
13. The cell line as claimed in claim 11, wherein the each of (A), (B), (C), (E), and (F), when present, are integrated into the genome of the cell line, and wherein the specified genes are expressed or expressible therefrom.
14. The cell line as claimed in claim 11, wherein the cells of the cell line are HEK293, HEK293T, HEK293A, PerC6 or 911 cells.
15. A process for producing an AAV cell line, the process comprising the steps of integrating into the genome of the cells of the cell line: (A) a nucleic acid molecule as claimed in claim 1, thereby producing a cell line that inducibly expresses viral cap and rep genes; and optionally one or more of the following: (B) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter or a minimal promoter comprising one or more cumate operator sites, operably-associated with (ii) a gene encoding a rc-CymR-TAD polypeptide; (C) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter or a constitutive promoter, operably-associated with (ii) a gene encoding a CymR; (D) a nucleic acid

molecule, vector or plasmid comprising: (i) one or more promoters, which may or may not comprise one or more cumate operator sites, operably-associated with (ii) one or more adenoviral genes which are competent to support AAV production.

16. A method of inducing transcription of cap and rep genes in a cell, the method comprising: inducing the transcription of cap and rep genes in a cell which comprises in its genome: (A) a nucleic acid molecule as claimed in claim 1, wherein the inducing is by the presence, in the cell, of a CymR-TAD polypeptide or by the expression or transient expression, in the cell, of a nucleic acid molecule which encodes a CymR-TAD polypeptide.

17. The method as claimed in claim 16, wherein the method additionally comprises the step of introducing into the cell: (B) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter or a constitutive promoter, operably-associated with (ii) a nucleotide sequence encoding a CymR-TAD; and/or (C) a nucleic acid molecule, vector or plasmid comprising: (i) one or more promoters, which may or may not comprise one or more cumate operator sites, operably-associated with (ii) one or more adenoviral genes which are competent to support AAV production; and/or (D) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter, operably-associated with (ii) a transgene, wherein the promoter and transgene are flanked by ITRs.

18. A method of inducing transcription of cap and rep genes in a cell, the method comprising: inducing the transcription of cap and rep genes in a cell which comprises: (A) a nucleic acid molecule as claimed in claim 1; (B) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter comprising one or more cumate operator sites, operably-associated with (ii) a gene encoding a rc-CymR-TAD polypeptide; and (C) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter, operably-associated with (ii) a gene encoding a CymR; and optionally (D) a nucleic acid molecule, vector or plasmid comprising: (i) one or more promoters, which may or may not comprise one or more cumate operator sites, operably-associated with (ii) one or more adenoviral genes which are competent to support AAV production; and/or (E) a nucleic acid molecule, vector or plasmid comprising: (i) a promoter, operably-associated with (ii) a transgene, wherein the promoter and transgene are flanked by ITRs, wherein the inducing is by the presence of or introduction into the cell of cumate.

19. The method of inducing transcription of cap and rep genes in a cell as claimed in claim 18, wherein (A), (B), (C), (D) and (E), when present, are all integrated into the genome of the cell.